ITNSA CISCO PROJECT

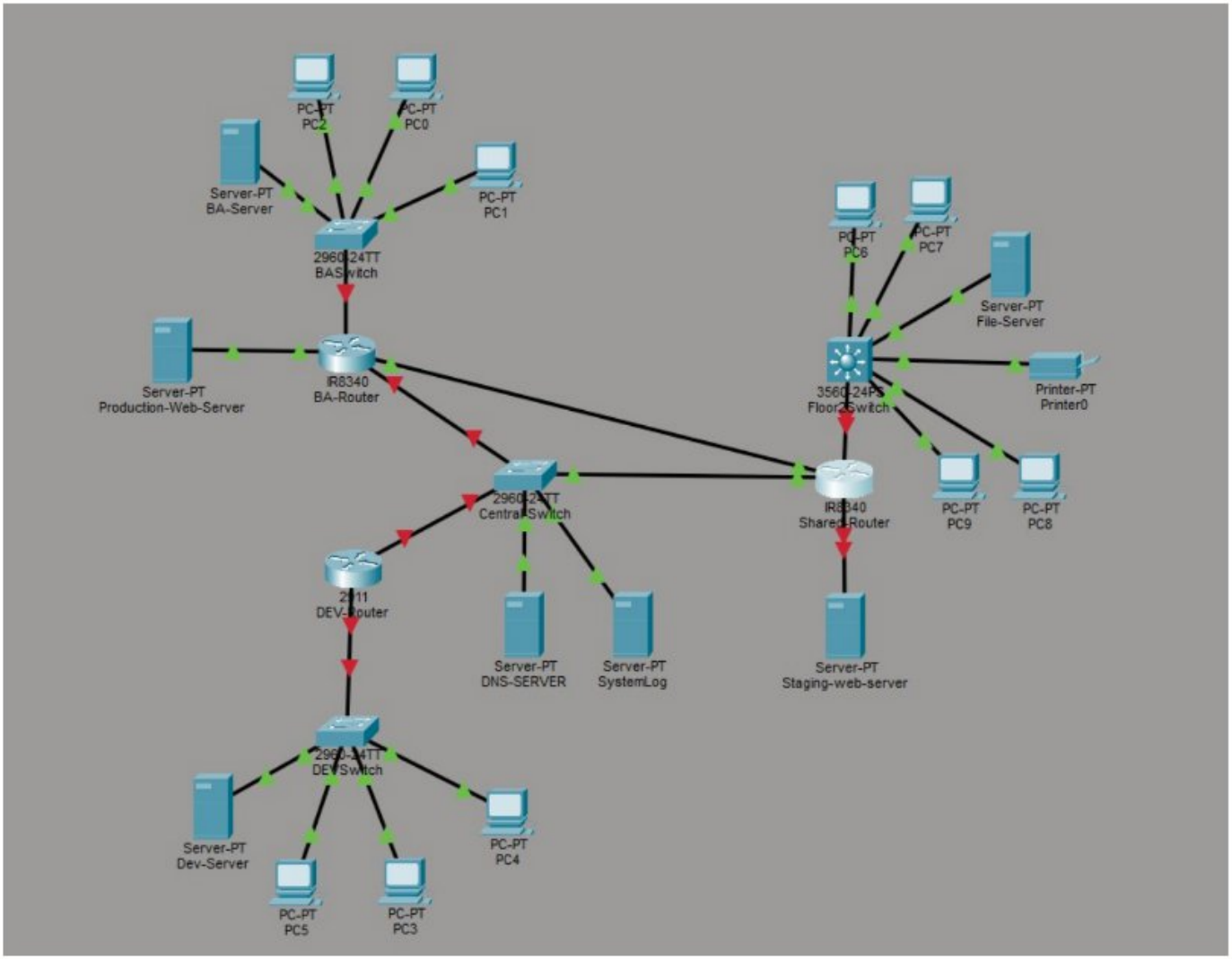
GROUP 1

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MR SIBONGINKOSI MAZIBUKO

ITNSA – Documentation:

From the scenario we can see that BrightSoft Solutions required a well-planned, segmented and secure network infrastructure across three dedicated floors. The design and configurations we have made in Cisco Packet Tracer ensures that each department has controlled access, reliable connectivity as well as isolated resources to the centralized servers.



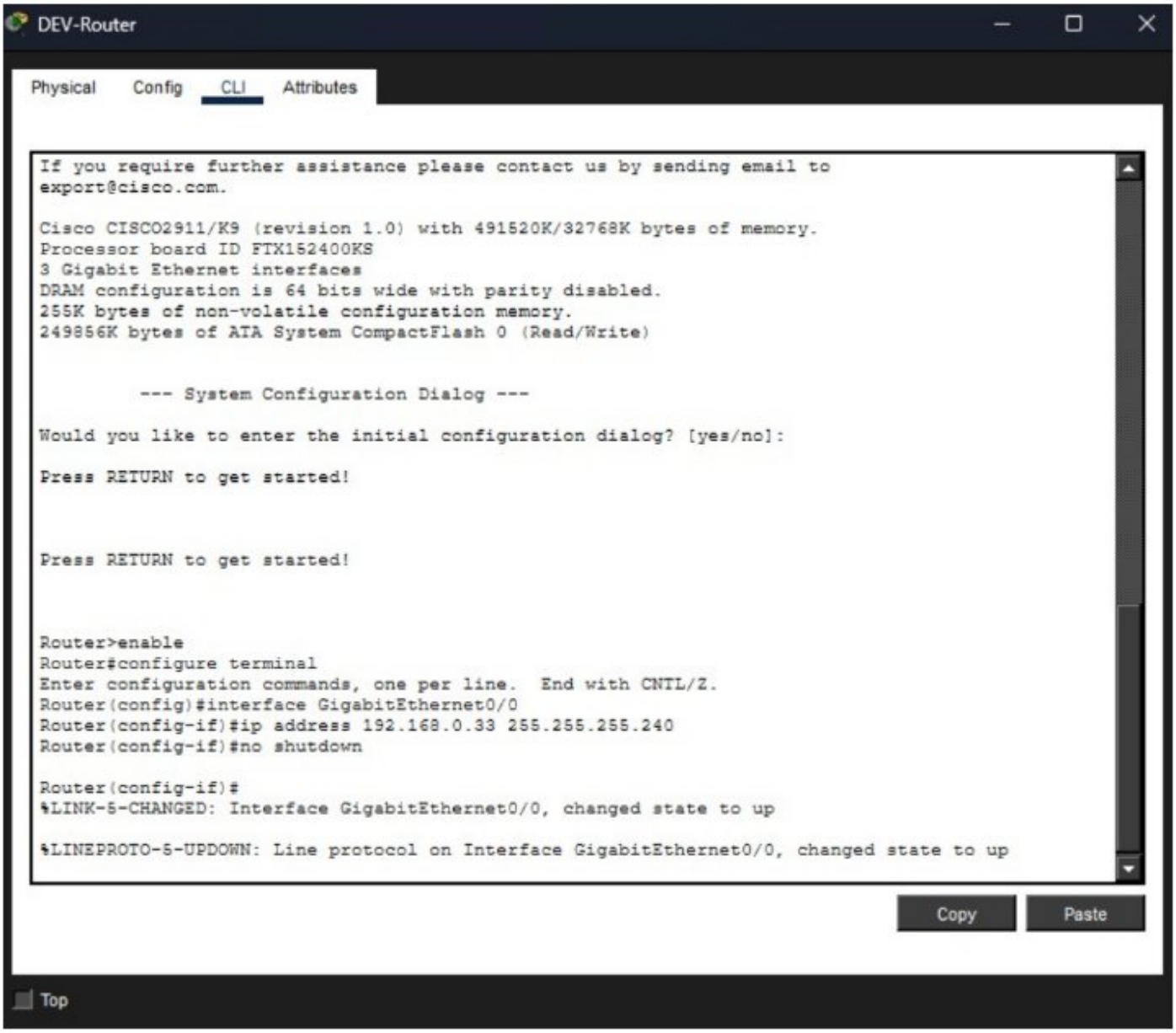
UnConfigured network layout

The picture above shows our unconfigured network layout that we will be using in this project with due understanding to the scenario and helping BrightSoft Solutions to resolve their problems. The above image includes the Developers, Business Analysts and Shared routers, switches, servers, PCs and a printer.

Pic – fully configured (paste final)

After thorough research and approval from other team members, we then started configuring our network in Cisco Packet Tracer and the above is the result of our fully configured network. Our fully configured network shows the different teams connected to their own respective routers, DHCP servers and centralized servers. We took this approach of segmenting by the floors and teams so it would prevent unnecessary traffic, improve the security and ensure that each department can be managed separately.

Developer's Router Configurations:



dev router configuration

The developer's router was configured from the CLI and brought online using no shutdown. It was assigned with the necessary IPs mentioned in the scenario. This provides the gateway for the Developer's subnet.

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 192.168.0.17 255.255.255.224
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

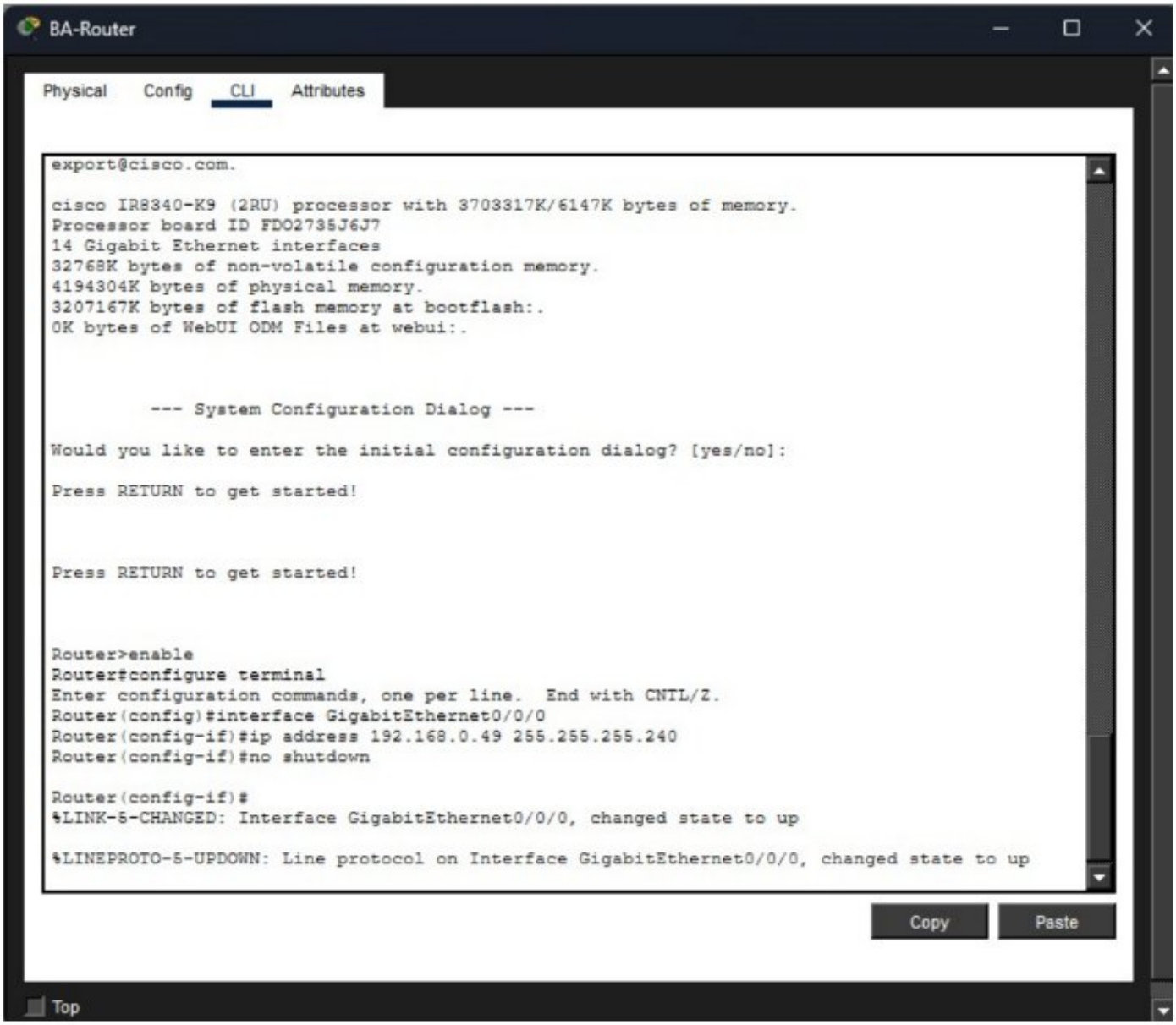
In the image above, the developer's router was connected to the central switch so that it can communicate with the other floors.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.0.0
Router(config-router)#end
Router#
```

dev router RIP

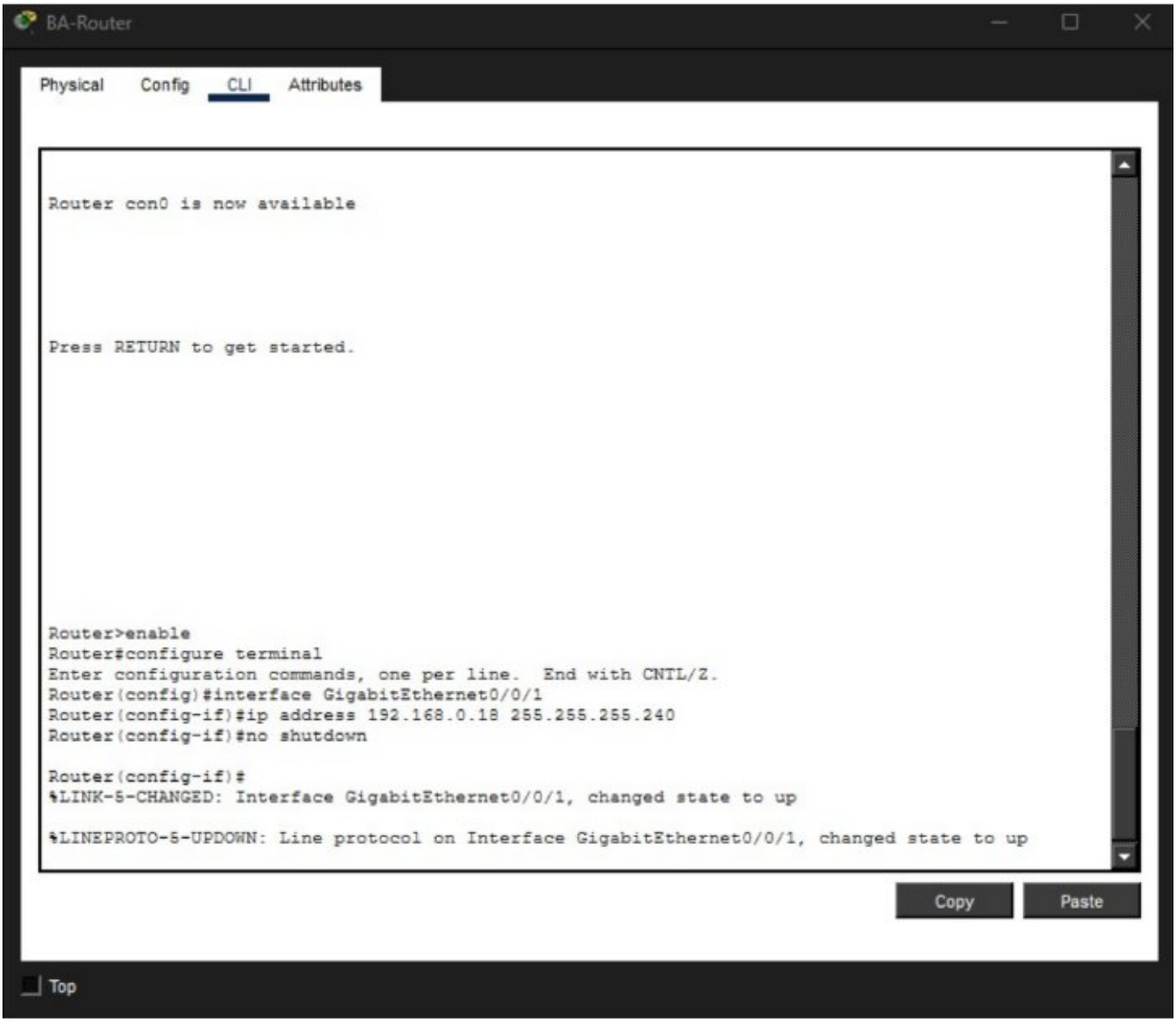
In the image above, we configured RIP v2 with the developer's router as it was required for the legacy applications mentioned in the scenario.

Business Analyst's Router Configurations:



BA router configuration

The business analyst's router was configured from the CLI and brought online using no shutdown. It was assigned with the necessary IPs mentioned in the scenario. This provides the gateway for the Business Analyst's subnet.



BA ROUTER TO CENTRAL SWITCH

In the image above, the business analyst's router was connected to the central switch so that it can communicate with the other floors.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.0.96 0.0.0.3 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

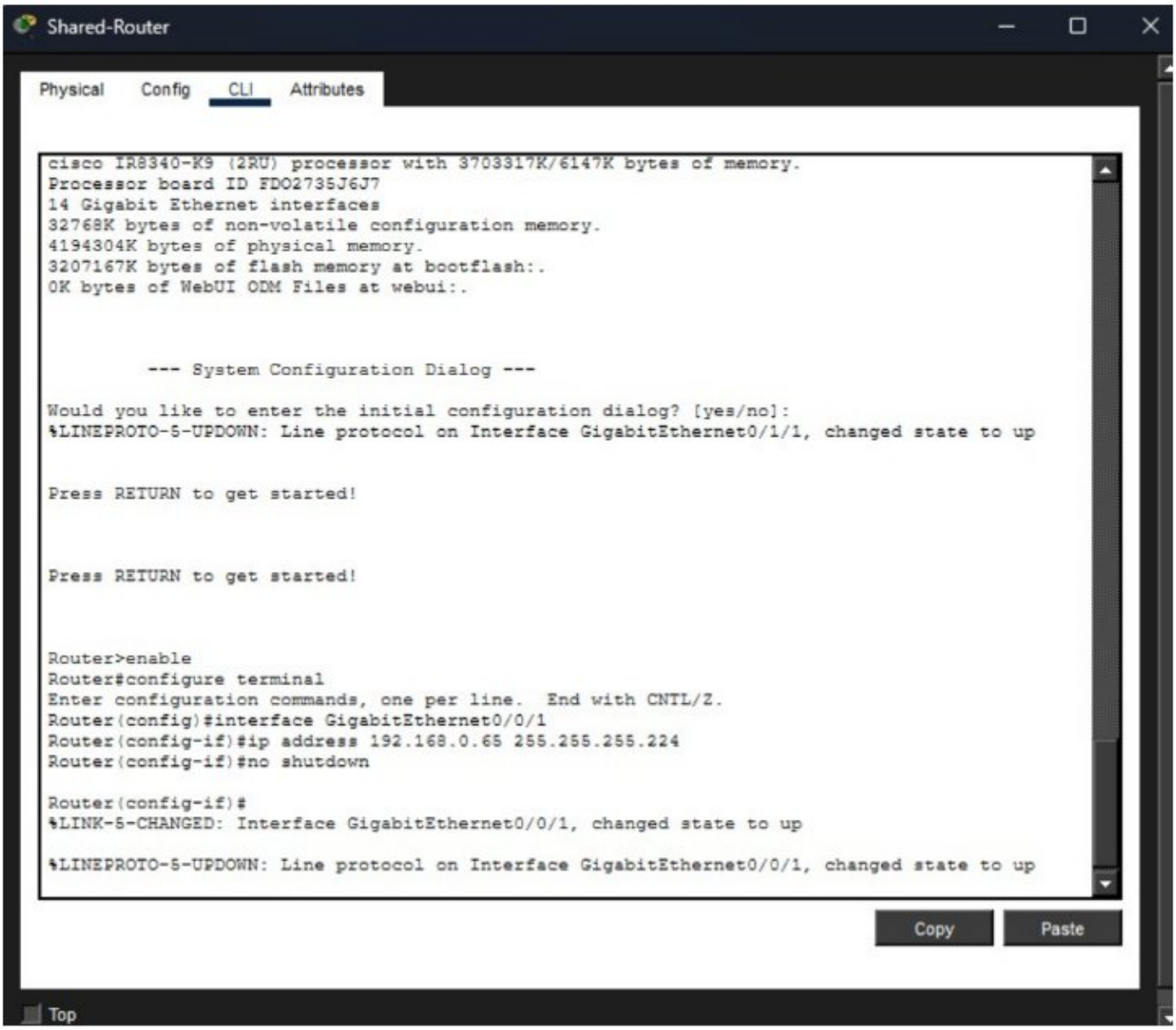
BA ROUTER OSPF

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#redistribute ospf 1 metric 1
Router(config-router)#router ospf 1
Router(config-router)#redistribute rip subnets
Router(config-router)#
```

BA ROUTER OSPF CONTINUED

In the images above, the OSPF and RIP was configured. The OSPF was chosen for scability as mentioned in the scenario.

Shared Router Configurations:



configuration for shared router

The shared router was configured from the CLI and brought online using no shutdown. It was assigned with the necessary IPs mentioned in the scenario. This connects Floor 2 and Floor 3 as well as providing internet access.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface vlan 10
Router(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

Router(config-if)#ip address 192.168.0.98 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface GigabitEtherent
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1/1 (1), with
Router GigabitEthernet0/1/1 (10).

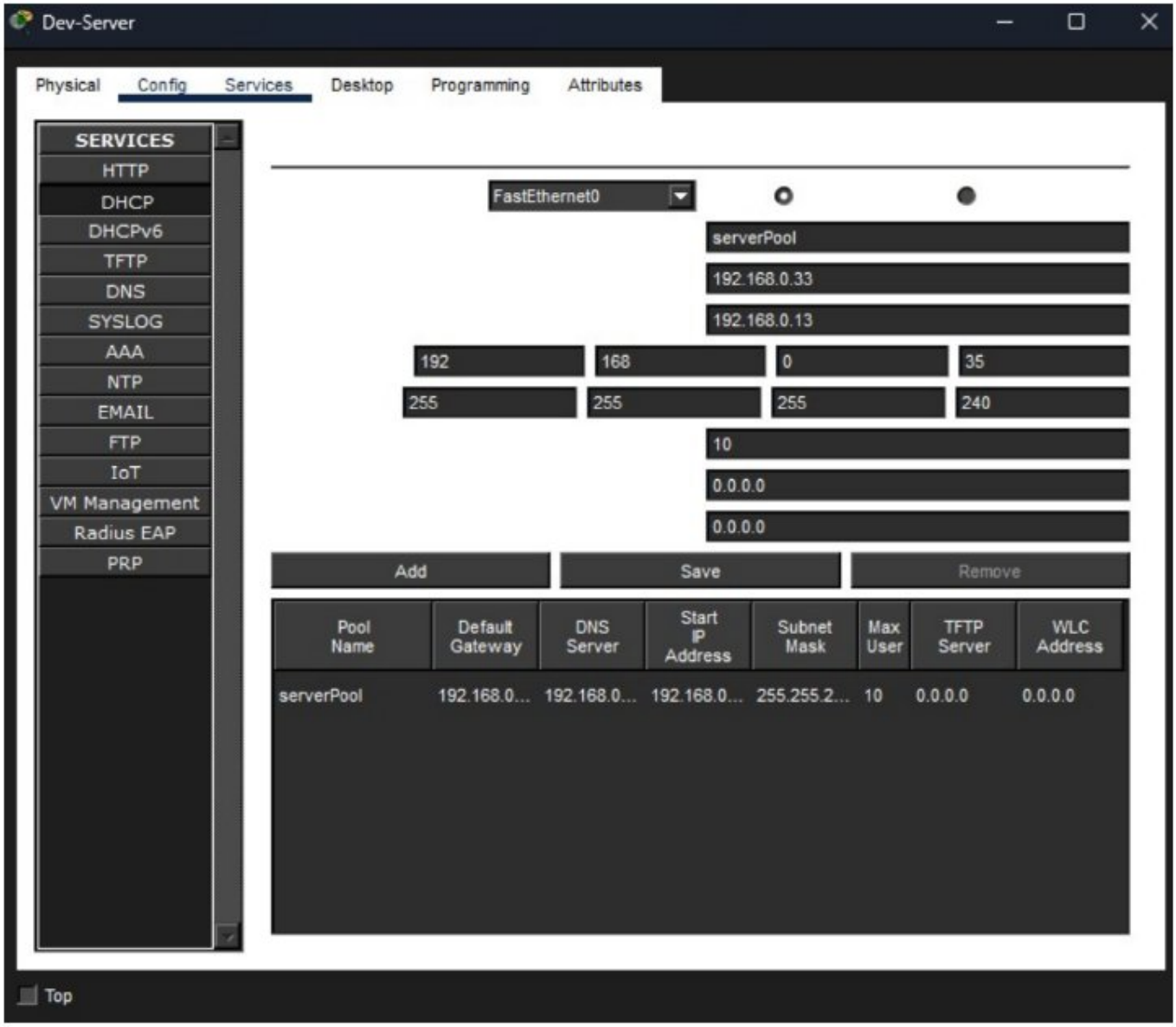
                                     ^
% Invalid input detected at '^' marker.

Router(config)#interface GigabitEthernet0/1/1
Router(config-if)#switchport mode access
Router(config-if)#switchport access vlan 10
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
exit
```

Shared router vlan 10 config

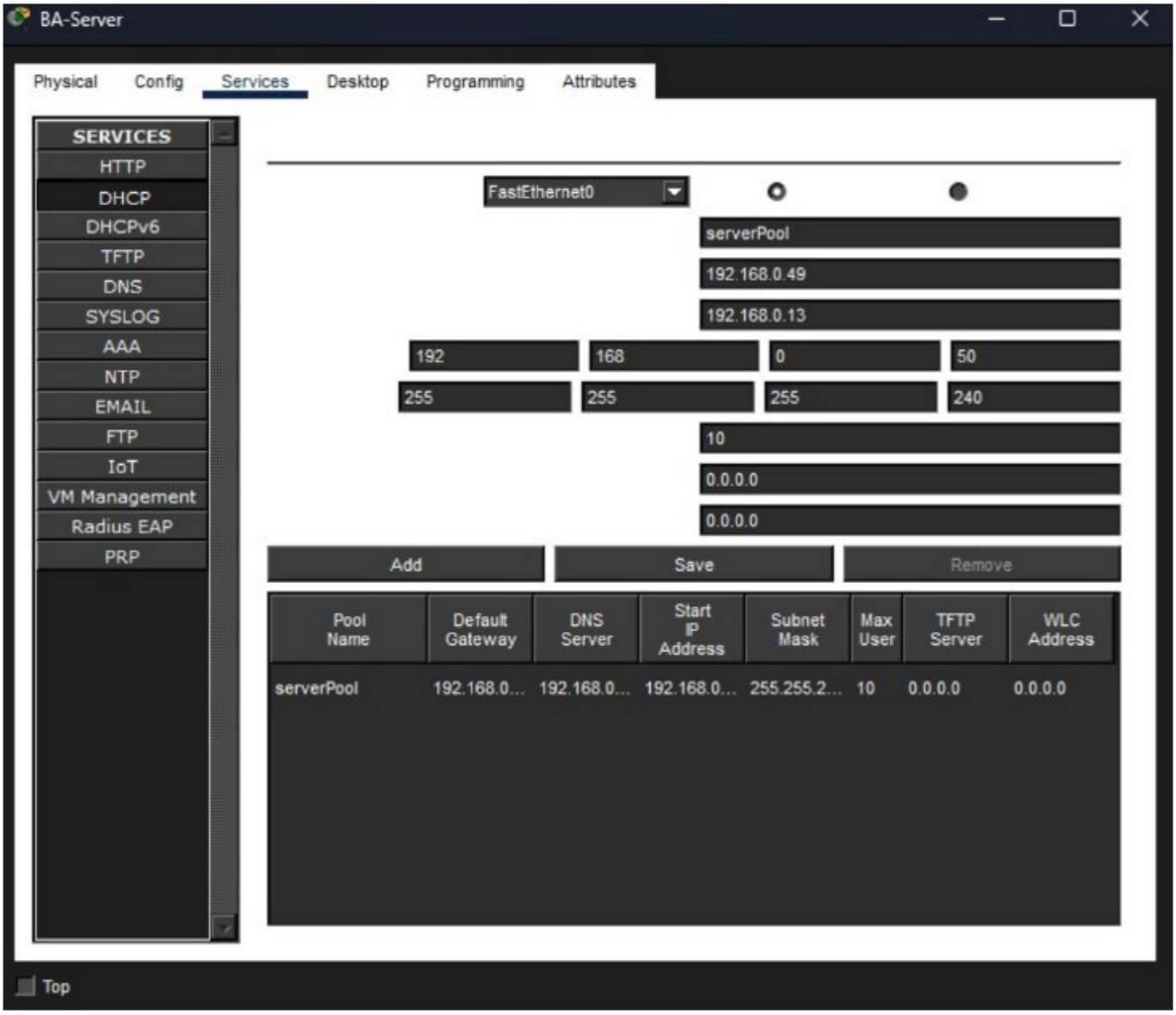
This assigning of the VLAN 10 separates the traffic of Floor 2 between the Test Engineers team and the UI Designers team into their own logical network. This ensures that their data remains secure which results in improved security and prevents each team from accessing it.

Developer’s DHCP Server:



DEV DHCP SERVER CONFIG

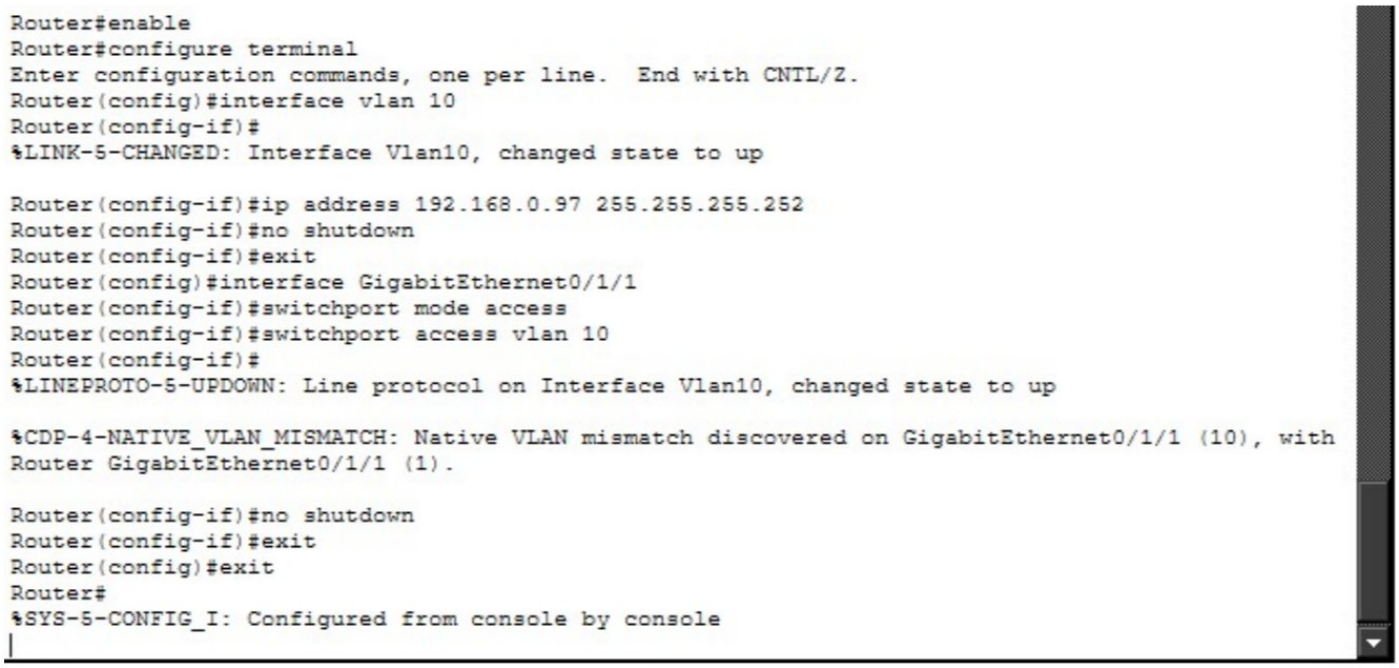
Business Analyst's DHCP Server:



BA DHCP SERVER CONFIG

The separate DHCP servers ensure that each team receives unique IP addresses only from its own subnet.

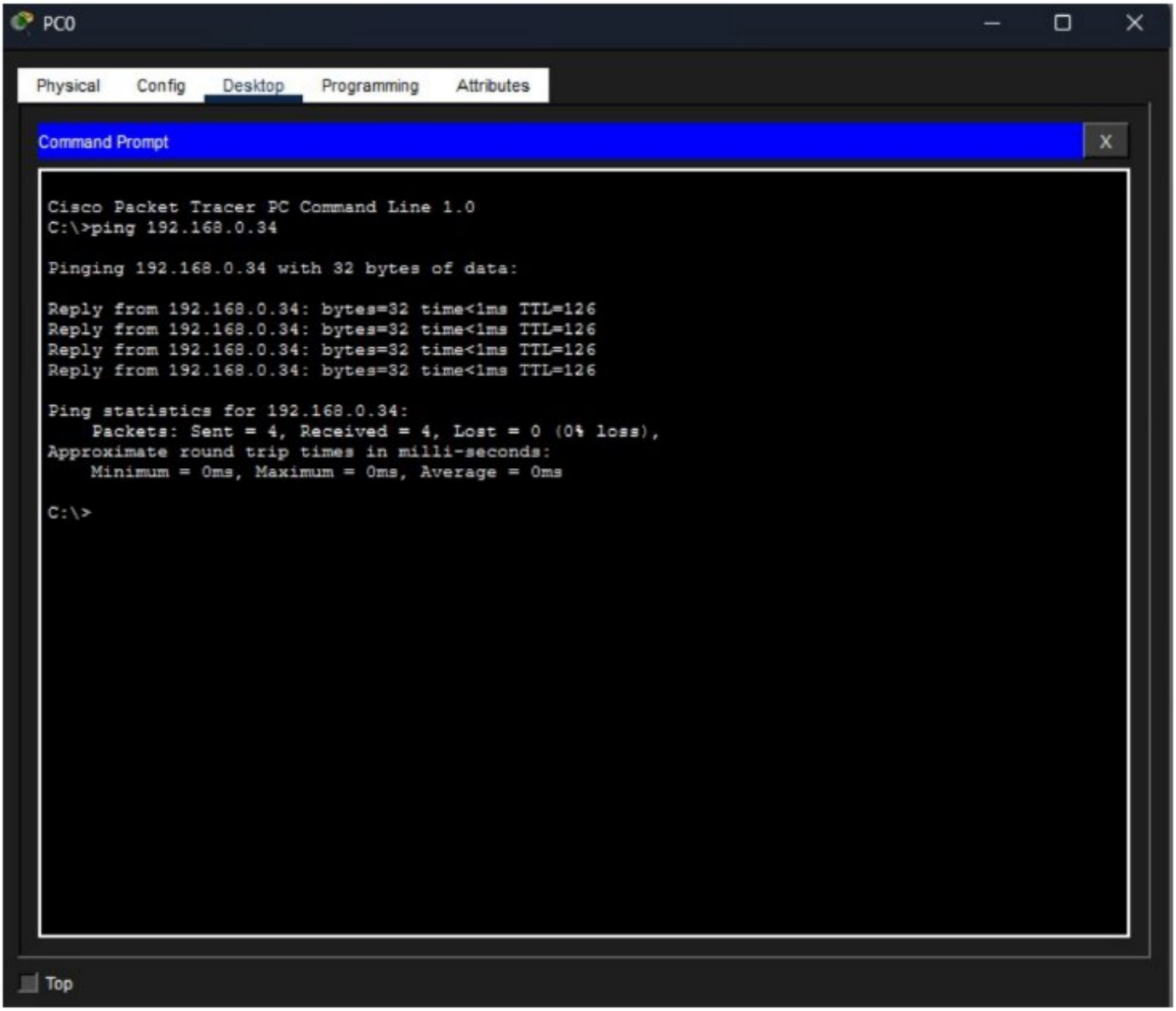
VLAN Configuration:



BA VLAN 10 CONFIG FOR 0/1/1

The VLAN Configuration separates the Test Engineers and the UI Designers on Floor 2 and thus improving the security as it prevents unauthorized access between the departments.

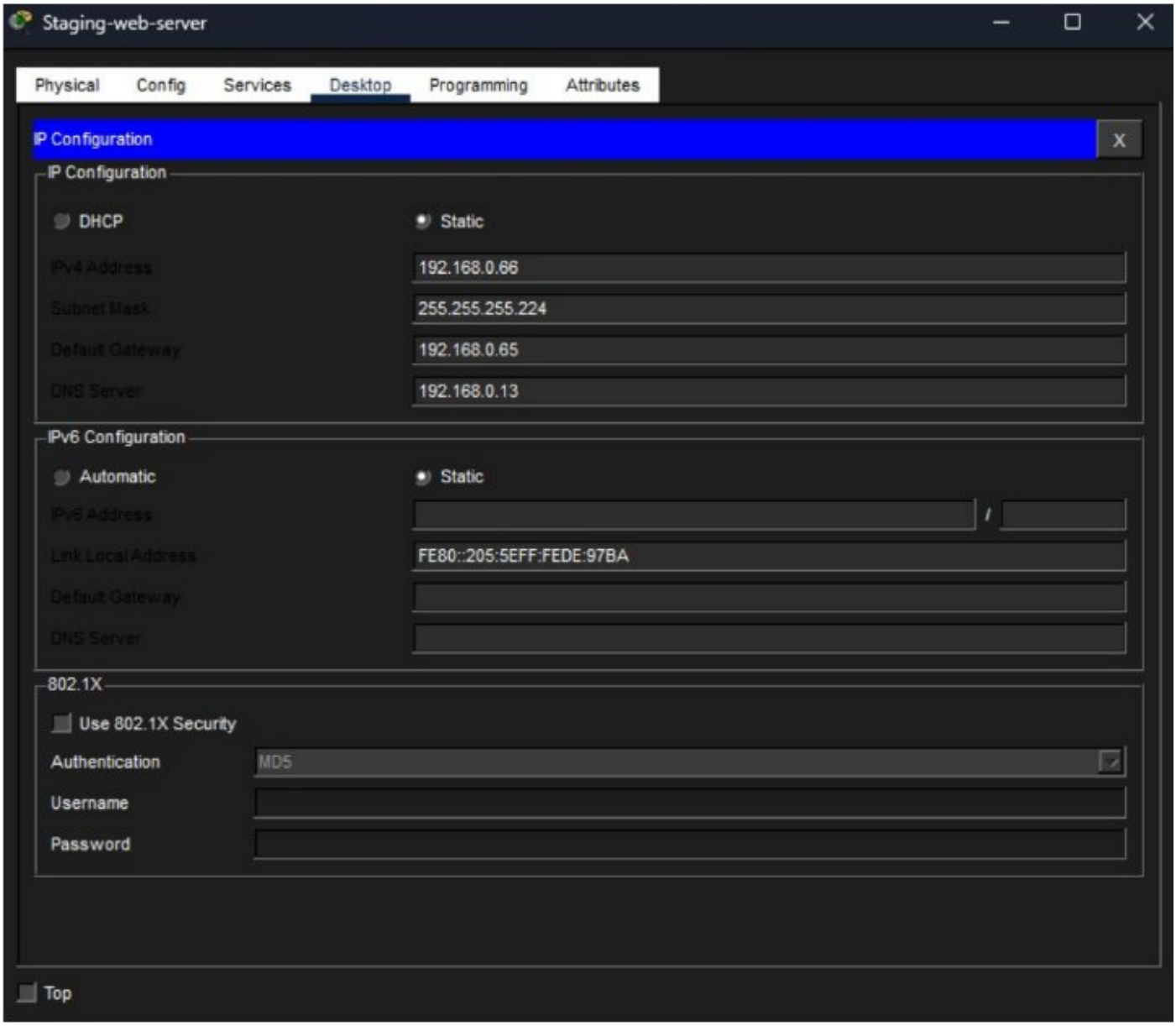
Testing:



proof BA can talk to dev

In the above image, a ping test was performed to verify the successful communication between the business analyst's and the developer's devices. The result proved correct routing and DHCP assignments as there were replies from all the packets sent.

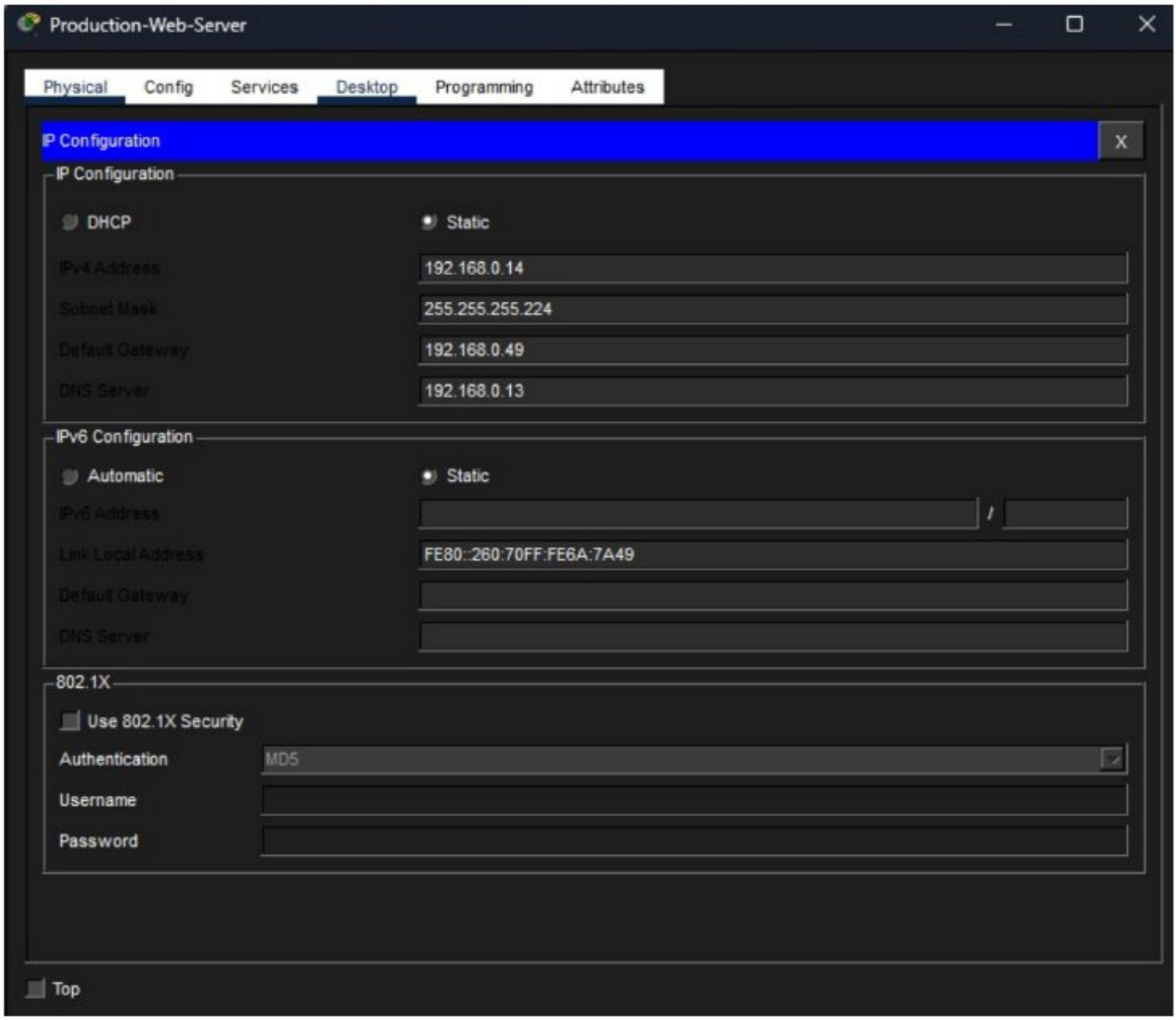
Staging Web Server:



staging web config

The above image shows the staging web server configurations. It is set up with a static IP address so that it can always be reached. It is accessible from the internet of HTTP/HTTPS.

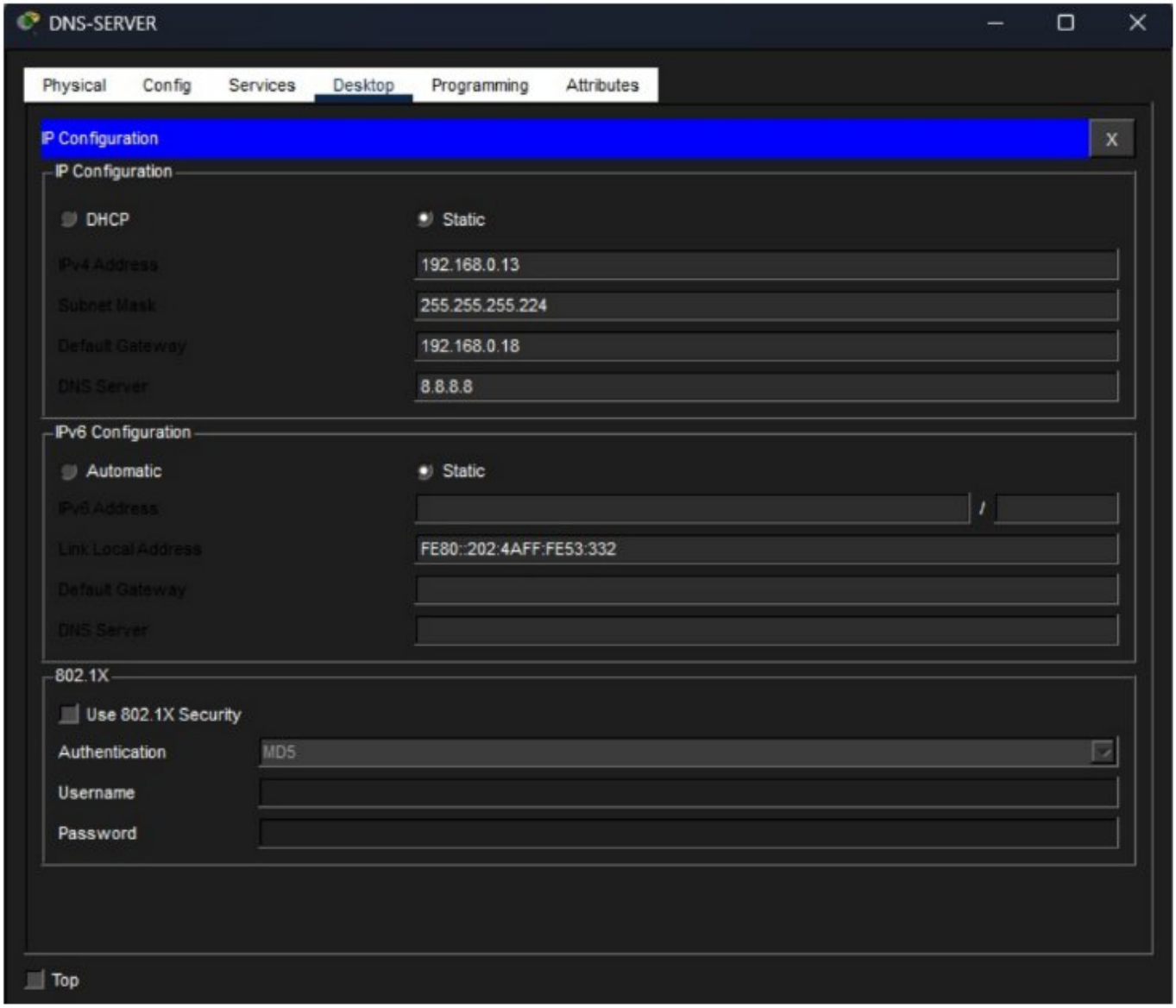
Production Web Server:



production config

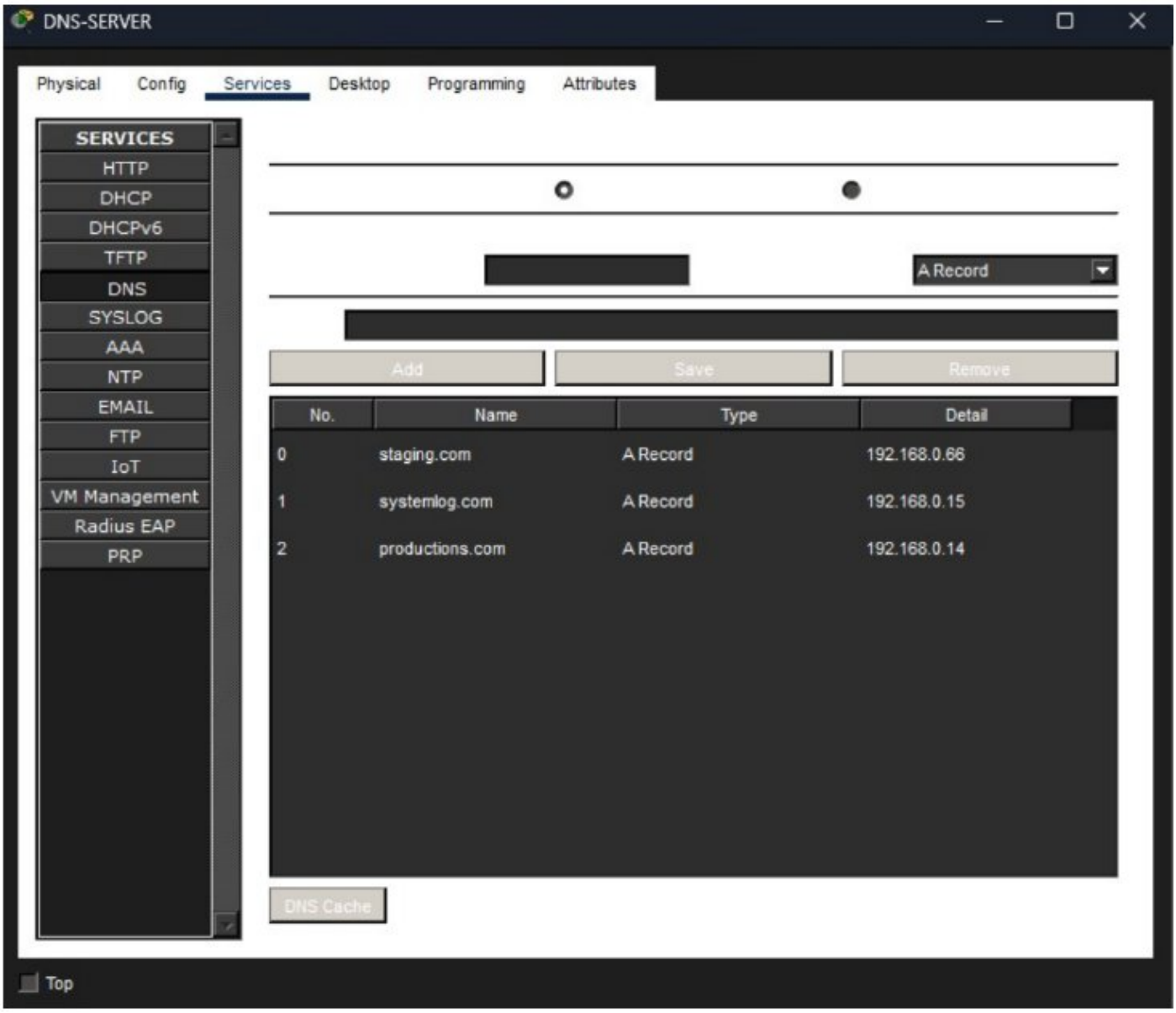
The above image shows the production web server configurations. It also given a static IP address for reliability and its accessible internally by all departments.

DNS Server:



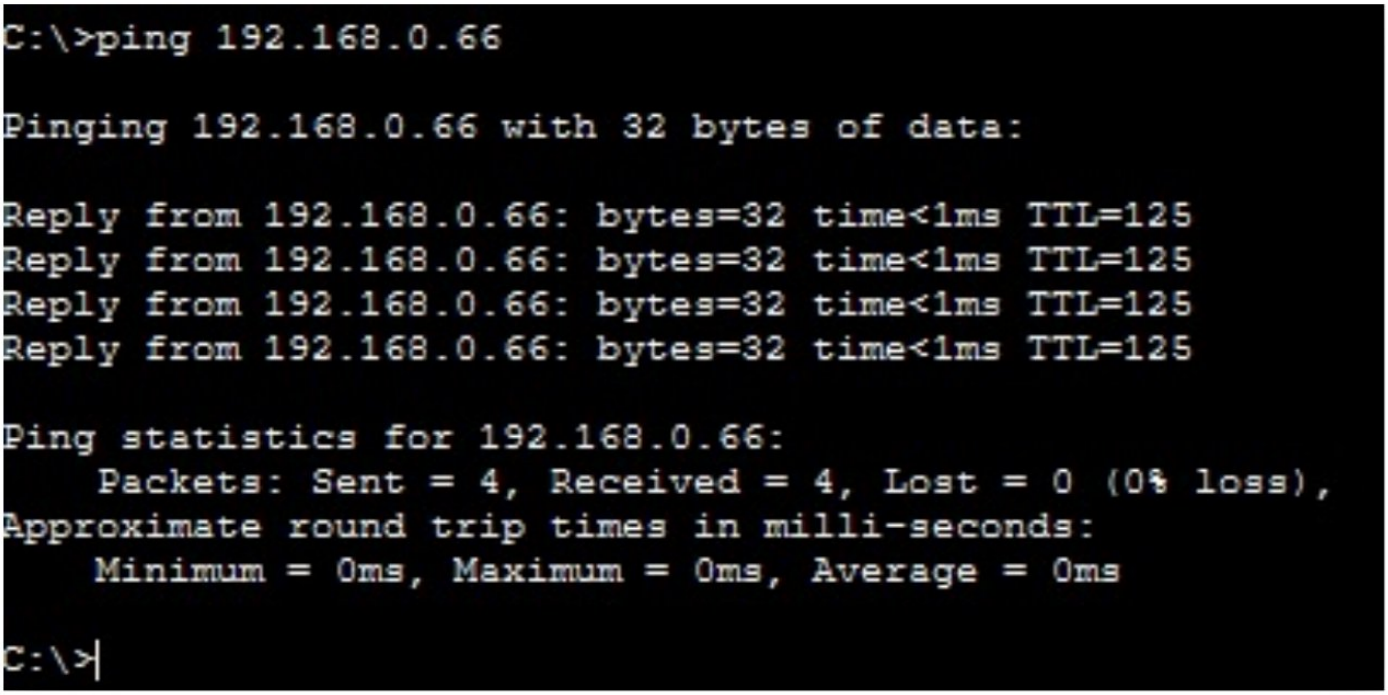
dns server configurations

The above image shows the DNS server configurations. It is configured with the DNS server as 8.8.8.8 which allows for searching on the internet.



dns services

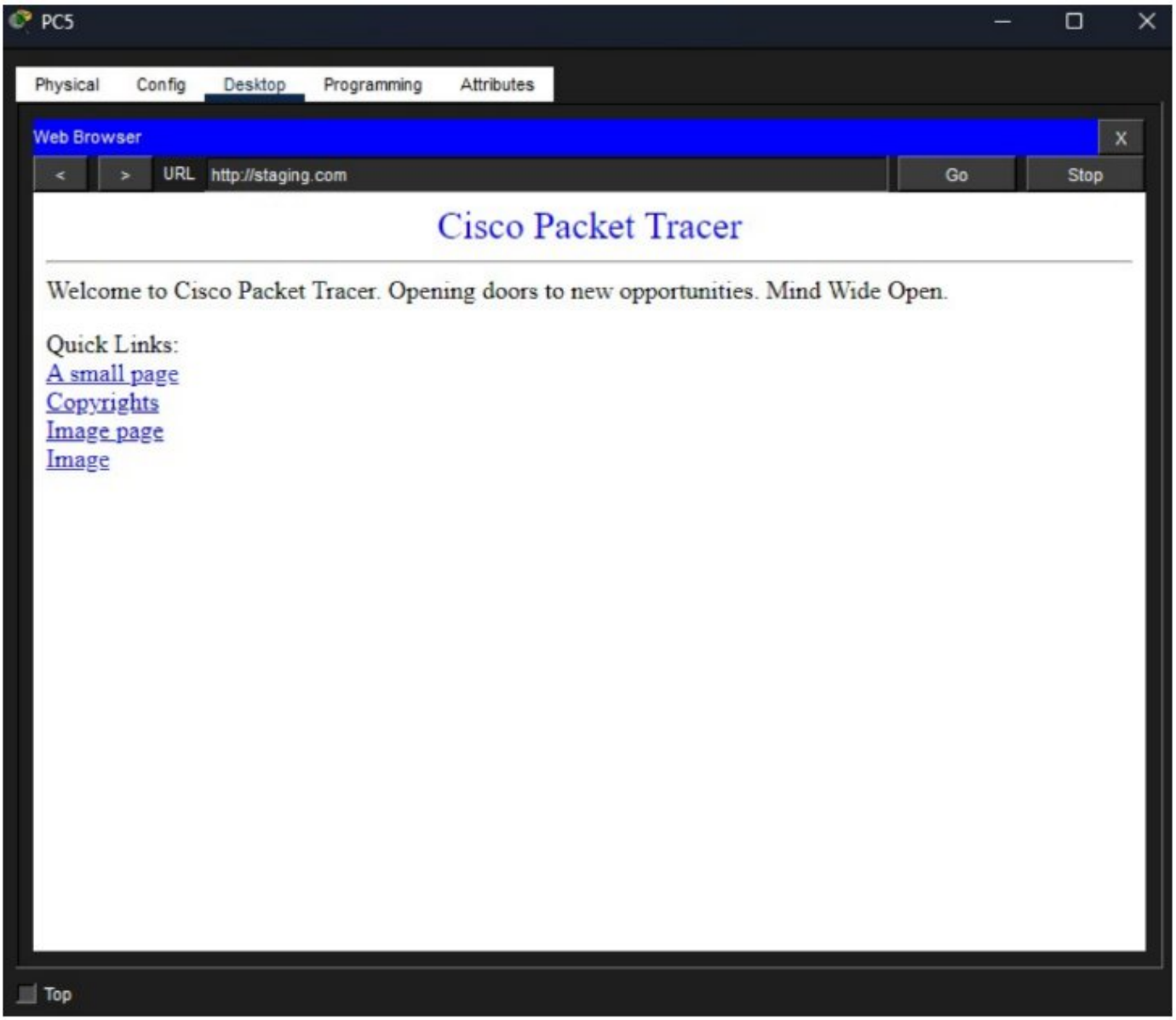
The image above shows the DNS being assigned to IP addresses for easier access as IP addresses would no longer need to be remembered and names can be used instead.



dev pc pinging the staging web server

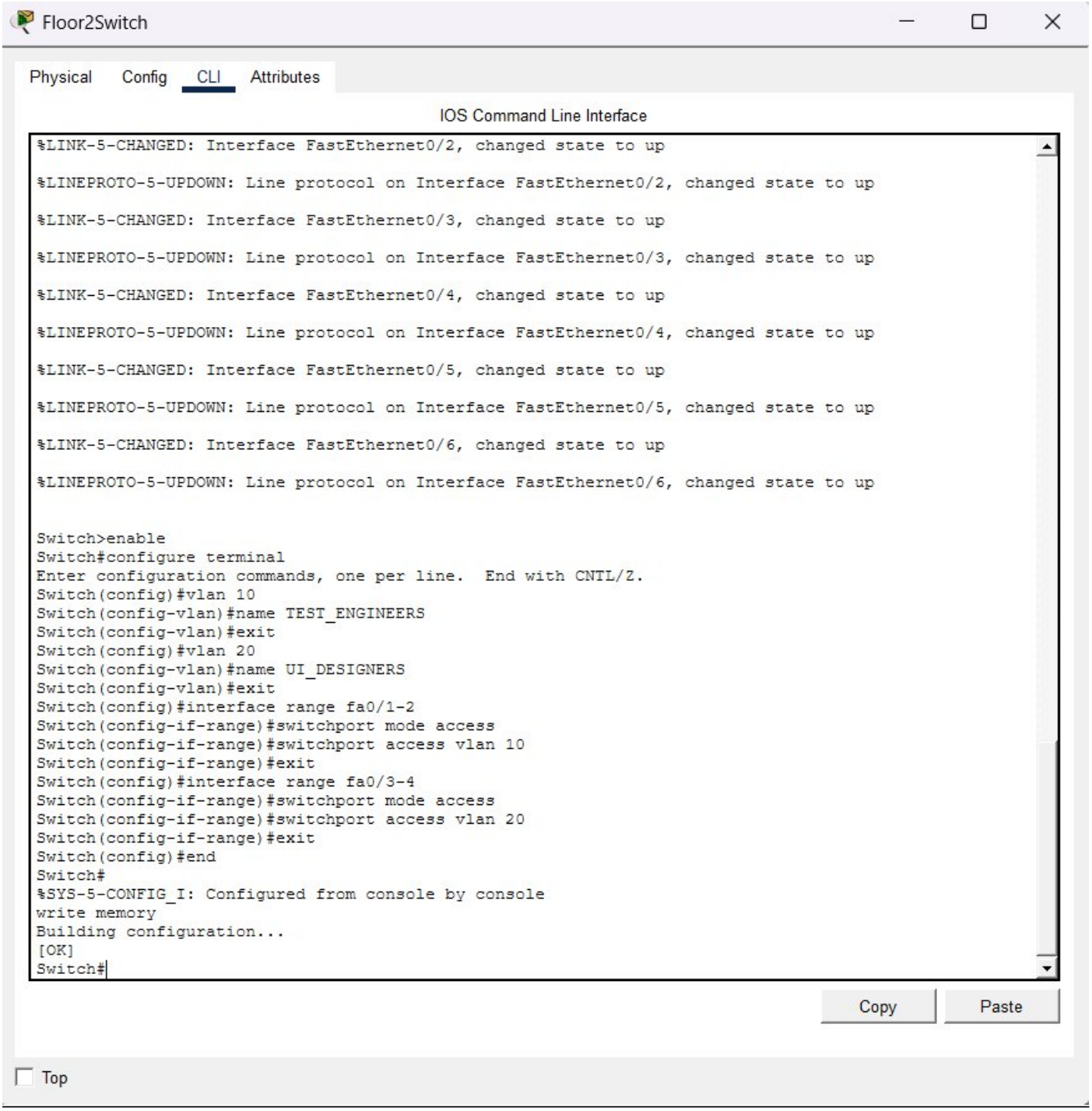
In the above image, we can see that the developer's pc is pinging the staging web server and all replies are successful indicating the connection between the two are correctly configured. The ping tests are performed to verify the basic IP connectivity.

As a result, below is the image which is testing the accessing of the web, and we can see that the developer's pc can access the staging web's page from the internet.



dev pc accessing the staging web from the internet

STORM START Q2 HERE



The above configurations show us the creation of the of two separate vlans. One was made for the test engineers and the other was made for the ui designers. The vlans are then switched to access mode which means that they are intended to connect to an end device (PC in this case). It also helps to connect the vlan to their ports.

```
VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                 Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                 Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                 Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                 Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                 Gig0/1, Gig0/2
10   TEST_ENGINEERS                    active    Fa0/1, Fa0/2
20   UI_DESIGNERS                      active    Fa0/3, Fa0/4
```

The above picture shows that the vlans and their names are active which means that they can be communicated to. As seen in the image above, these vlans are connected via fa0/1 to fa0/4.

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#ip routing
Switch(config)#interface vlan 10
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

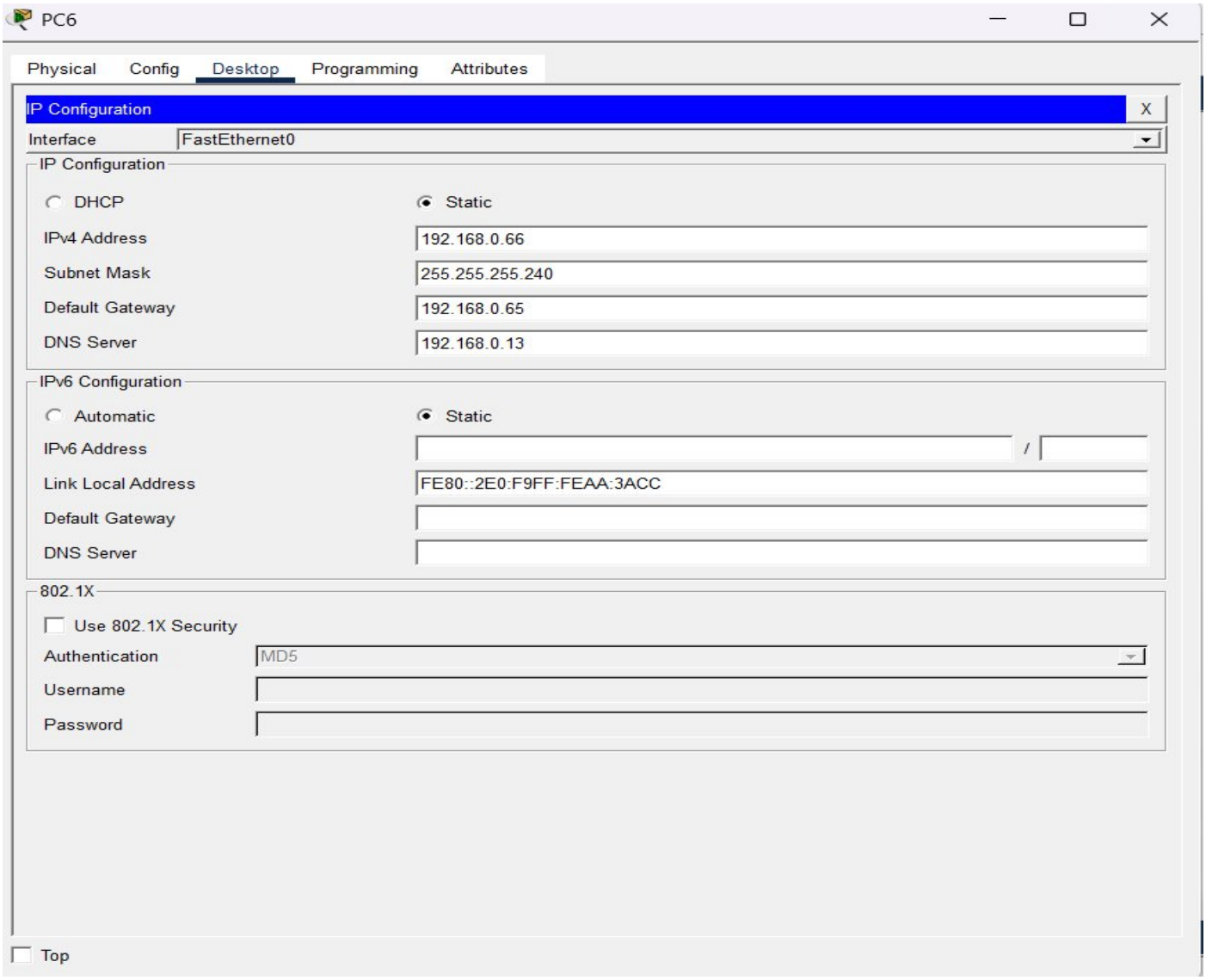
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up
ip address 192.168.0.97
% Incomplete command.
Switch(config-if)#ip address 192.168.0.97
% Incomplete command.
Switch(config-if)#ip address 192.168.0.97 255.255.255.240
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

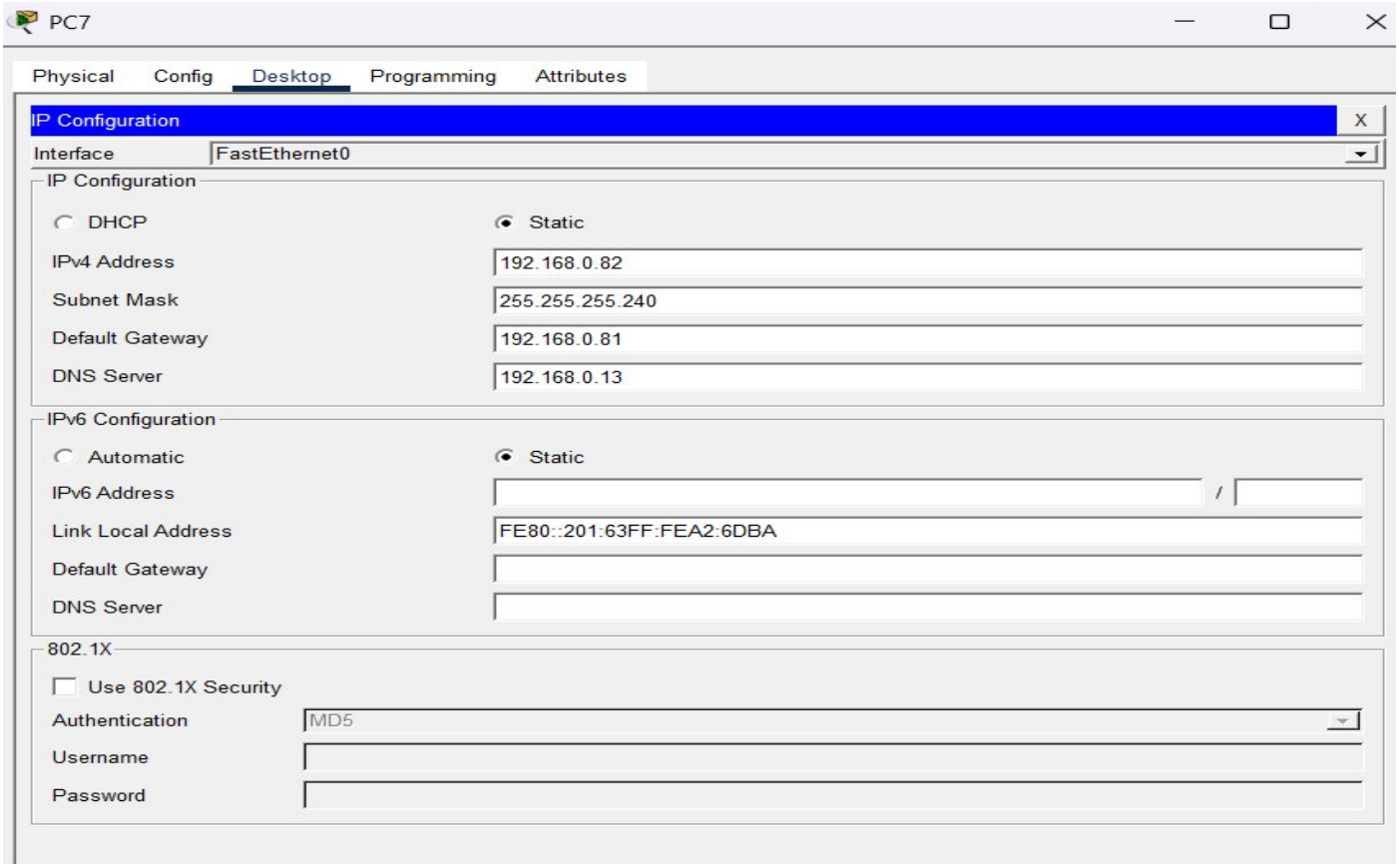
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

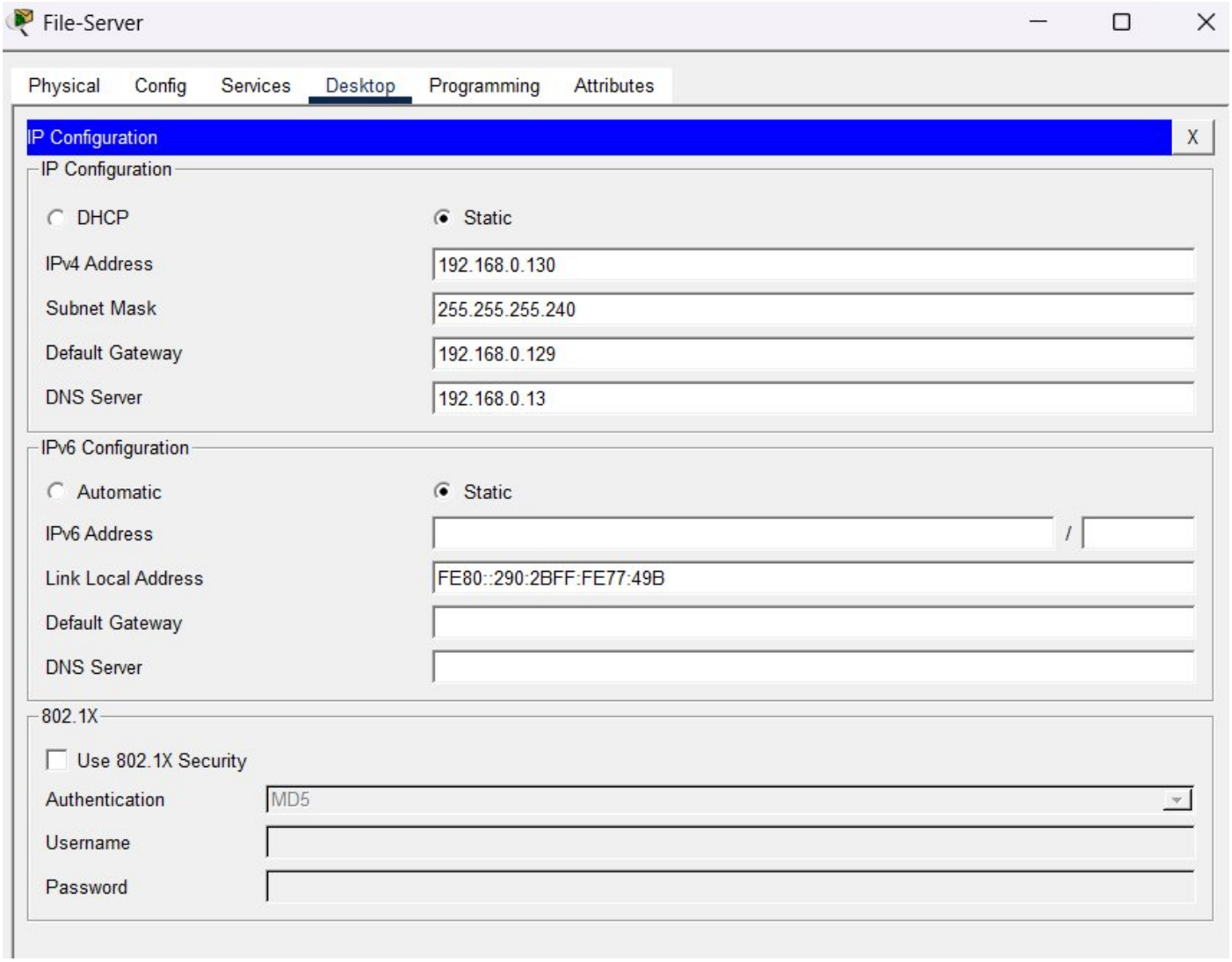
Switch(config-if)#ip address 192.168.0.113 255.255.255.240
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface g0/1
Switch(config-if)#no switchport
Switch(config-if)#ip address 192.168.0.34 255.255.255.240
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#ip route 0.0.0.0 0.0.0.0 192.168.0.35
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Switch#
```

Copy Paste

The above picture shows us the ip addresses being assigned through the CLI. This ensures that there is an address that can be used to communicate. It means that resources have a destination to go to when they are requested.


PC6
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.0.66
Subnet Mask
255.255.255.240
Default Gateway
192.168.0.65
DNS Server
192.168.0.13
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::2E0:F9FF:FEAA:3ACC
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password
Top

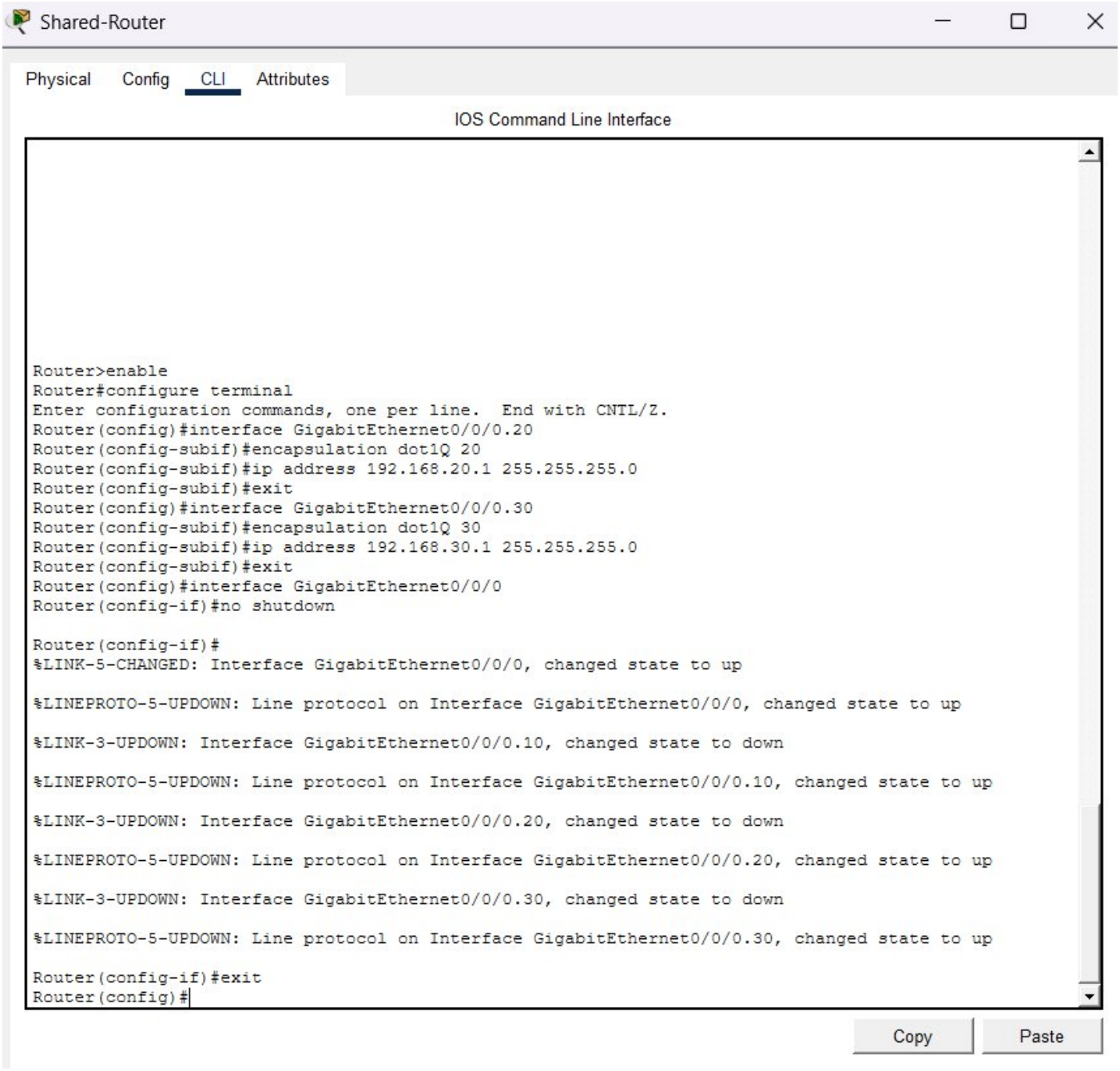

PC7
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.0.82
Subnet Mask
255.255.255.240
Default Gateway
192.168.0.81
DNS Server
192.168.0.13
IPv6 Configuration
Automatic
Static
IPv6 Address
/
Link Local Address
FE80::201:63FF:FEA2:6DBA
Default Gateway
DNS Server
802.1X
Use 802.1X Security
Authentication
MD5
Username
Password



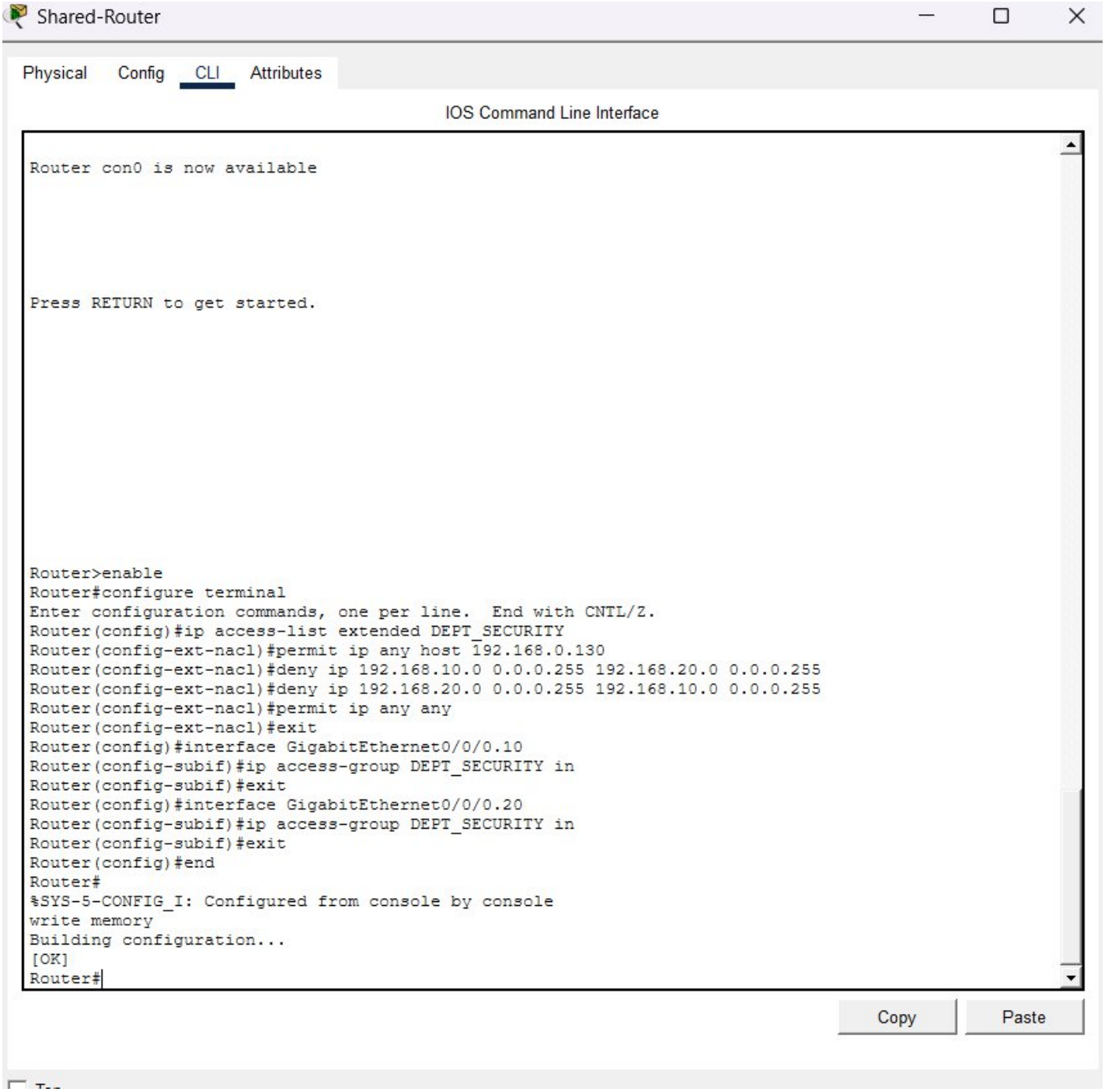
The 3 images above are the configurations for PC6, PC7 and the File Server. These configurations were needed in question 2.2 where the layer 3 switch had to be configured.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0.10
Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.10.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

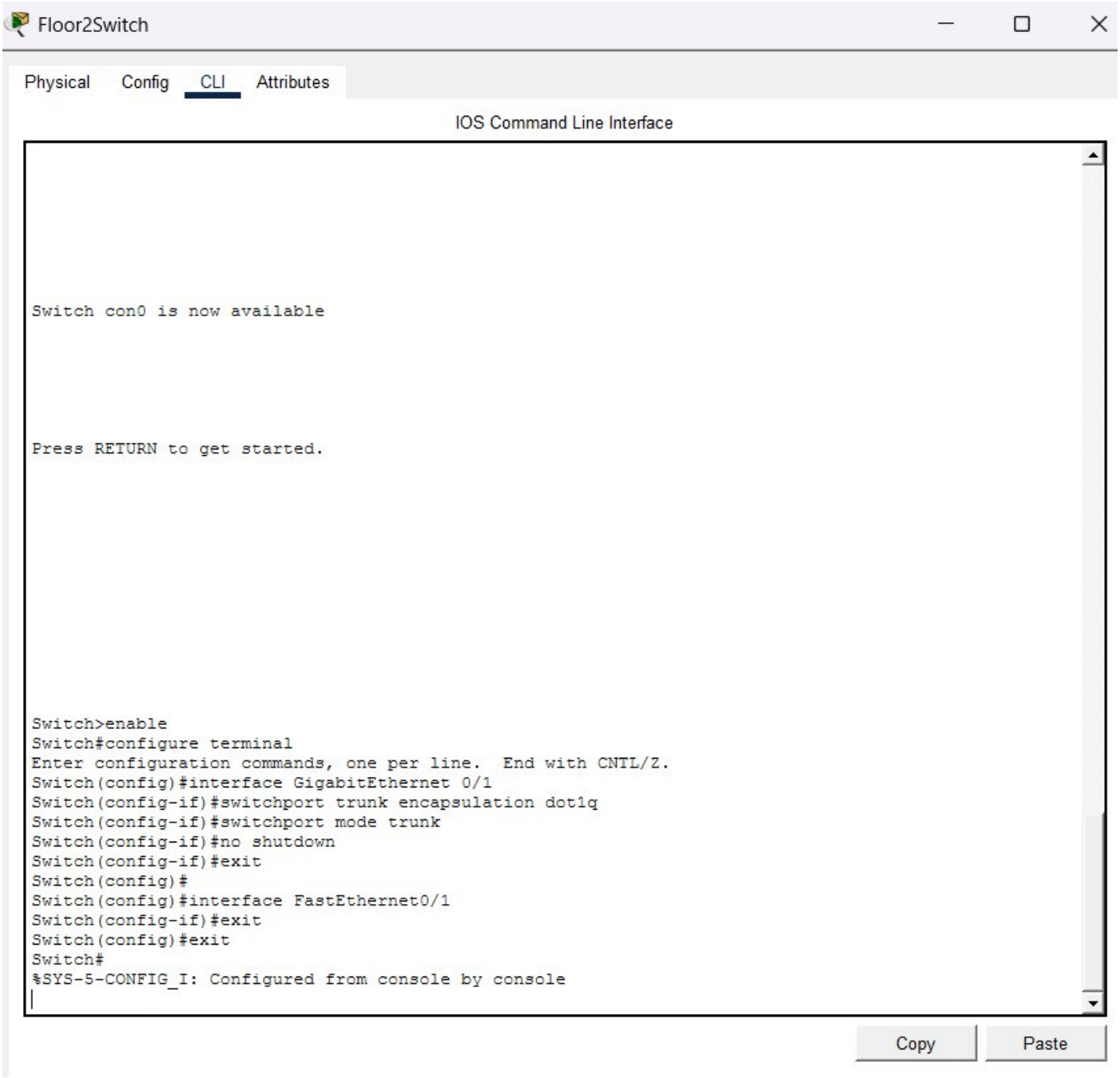
This is the creation of the sub interfaces that were needed for question 2.2. It also shows the encapsulation that takes place within vlan 10. The encapsulation helps with a few things. Firstly, it helps to identify the owner of the data itself by attaching a tag to it when it leaves the source. It also helps us to match the router to the switch. Most importantly, it enables inter vlan routing. This means that traffic can move between subnets without issue.



The image above shows the the other 2 vlans having their sub interfaces created and shows the encapsulation in both of them as well. It works the same as the previous configurations and are done for the same purpose as well.



The company also wanted another level of security. This is where ACL comes in. The ACL gives an extended layer of security. In the image above we can see that some IP addresses are allowed access to resources while some have been denied. This ensures that the principal of least privilege is followed and that groups are only allowed access to resources that they require.



The above image shows the trunking that has to take place within the network. The trunking allows a single cable to carry the traffic of multiple vlans simultaneously. This not only reduces cable clutter but also uses less hardware ports so that they can be used for other cabling.

```
Switch#show interface trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/7       on           802.1q         trunking      1

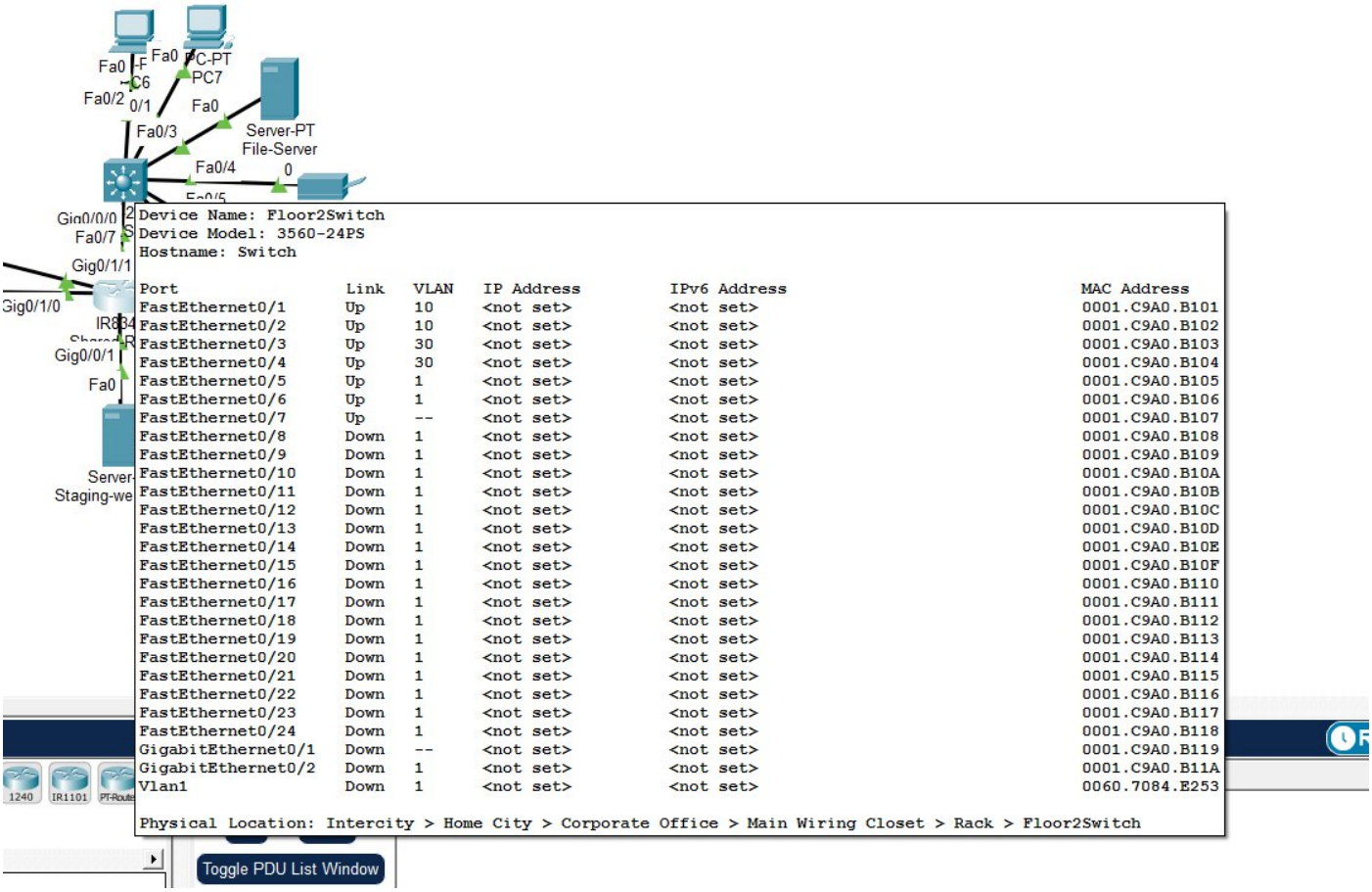
Port        Vlans allowed on trunk
Fa0/7       2-1001

Port        Vlans allowed and active in management domain
Fa0/7       10,20,30

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/7       10,20,30

Switch#
```

This image shows proof of the trunking. We can see that trunking is allowed on the vlans.



The above image shows the vlan assignment to the fast ethernet.

PAARTH START Q3 HERE

The main purpose of this question was to configure the staging environment so that approved external partners can access the staging web service over HTTP/HTTPS while maintaining strict isolation from the rest of the internal network.

```
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#show ip interface brief | include GigabitEthernet0/0/0
%SYS-5-CONFIG_I: Configured from console by console

GigabitEthernet0/0/0   unassigned      YES unset  down                  down
GigabitEthernet0/0/0.10192.168.10.1    YES manual up                    down
GigabitEthernet0/0/0.20192.168.20.1    YES manual up                    down
GigabitEthernet0/0/0.30192.168.30.1    YES manual up                    down
GigabitEthernet0/0/0.40192.168.3.1     YES manual up                    down
Router#show ip interface brief | include GigabitEthernet0/0/0
GigabitEthernet0/0/0   unassigned      YES unset  down                  down
GigabitEthernet0/0/0.10192.168.10.1    YES manual up                    down
GigabitEthernet0/0/0.20192.168.20.1    YES manual up                    down
GigabitEthernet0/0/0.30192.168.30.1    YES manual up                    down
GigabitEthernet0/0/0.40192.168.3.1     YES manual up                    down
Router#enable
Router#show ip interface brief | include GigabitEthernet0/0/0
GigabitEthernet0/0/0   unassigned      YES unset  down                  down
GigabitEthernet0/0/0.10192.168.10.1    YES manual up                    down
GigabitEthernet0/0/0.20192.168.20.1    YES manual up                    down
GigabitEthernet0/0/0.30192.168.30.1    YES manual up                    down
GigabitEthernet0/0/0.40192.168.3.1     YES manual up                    down
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1/2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.30, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0.40, changed state to up

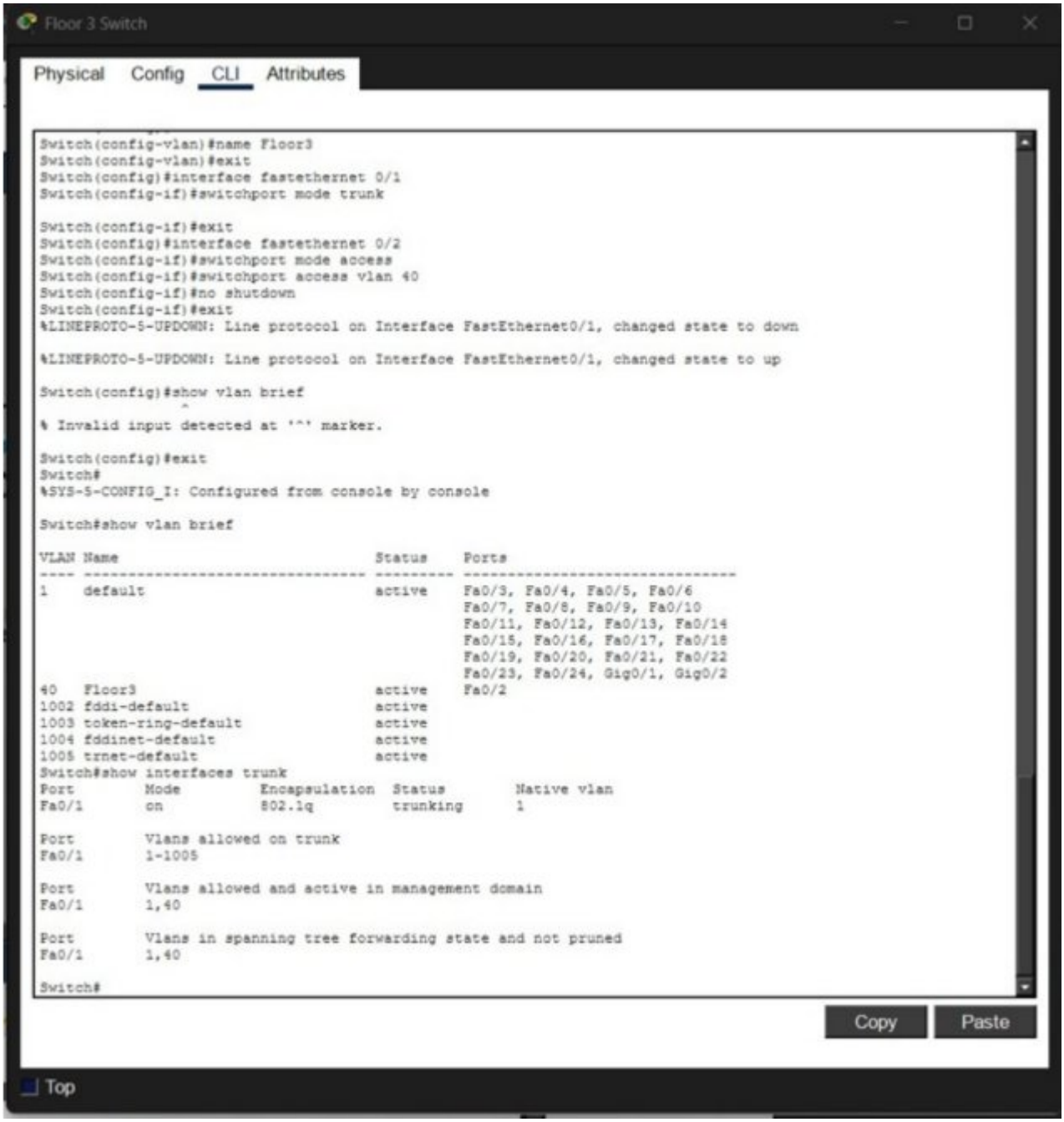
Router#show ip interface brief | include 0/0/0
GigabitEthernet0/0/0   unassigned      YES unset  up                    up
GigabitEthernet0/0/0.10192.168.10.1    YES manual up                    up
GigabitEthernet0/0/0.20192.168.20.1    YES manual up                    up
GigabitEthernet0/0/0.30192.168.30.1    YES manual up                    up
GigabitEthernet0/0/0.40192.168.3.1     YES manual up                    up
Router#ping 192.168.3.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.10, timeout is 2 seconds:
..
```

Sub-interfaces were created and each of them were assigned an IP address. The command “show ip interface brief” was to confirm that all interfaces were up and assigned correctly. The sub-interfaces with the VLAN IDs allowed segmentation and the separation of traffic between the different teams. This also helped with restricting lateral movement between the different teams.



To maintain strict isolation from the other teams it was essential that the staging environment was placed in its own VLAN. VLAN 40 was assigned to the staging environment on floor 3. This ensures that only approved external partners and the developer's team could reach the staging web service.



Trunking was required to carry multiple VLAN's across a single physical link which also ensured scalability. It was also essential to verify that the VLAN 40 was created and active so traffic could pass through the trunk and also ensure that the staging environment is securely isolated but still be reachable for controlled traffic.



The shared router needed to be configured with sub-interfaces for multiple VLAN’s to allow for inter-VLAN routing and provide a connection between the teams and the staging environment. Before the ACL’s and NAT was applied, we had to ensure that the staging server was reachable. The command “show ip interface brief” was used to confirm that all sub-interfaces were up and assigned correctly.



The NAT (Network Address Translation) allows the staging server to be reachable from the Internet whereas the ACLs (Access control list) ensures that only specific traffic is allowed. This prevents unauthorized access or lateral movement into other VLANs. The configurations ensure that approved external partners can reach the staging server with approved protocols only. It also restricts the teams from accessing VLAN 40. The Development team can update the staging server, but no other team can reach it. Furthermore, NAT hides the IP addresses, the ACLs block traffic and the VLAN segmentation isolates the different departments.



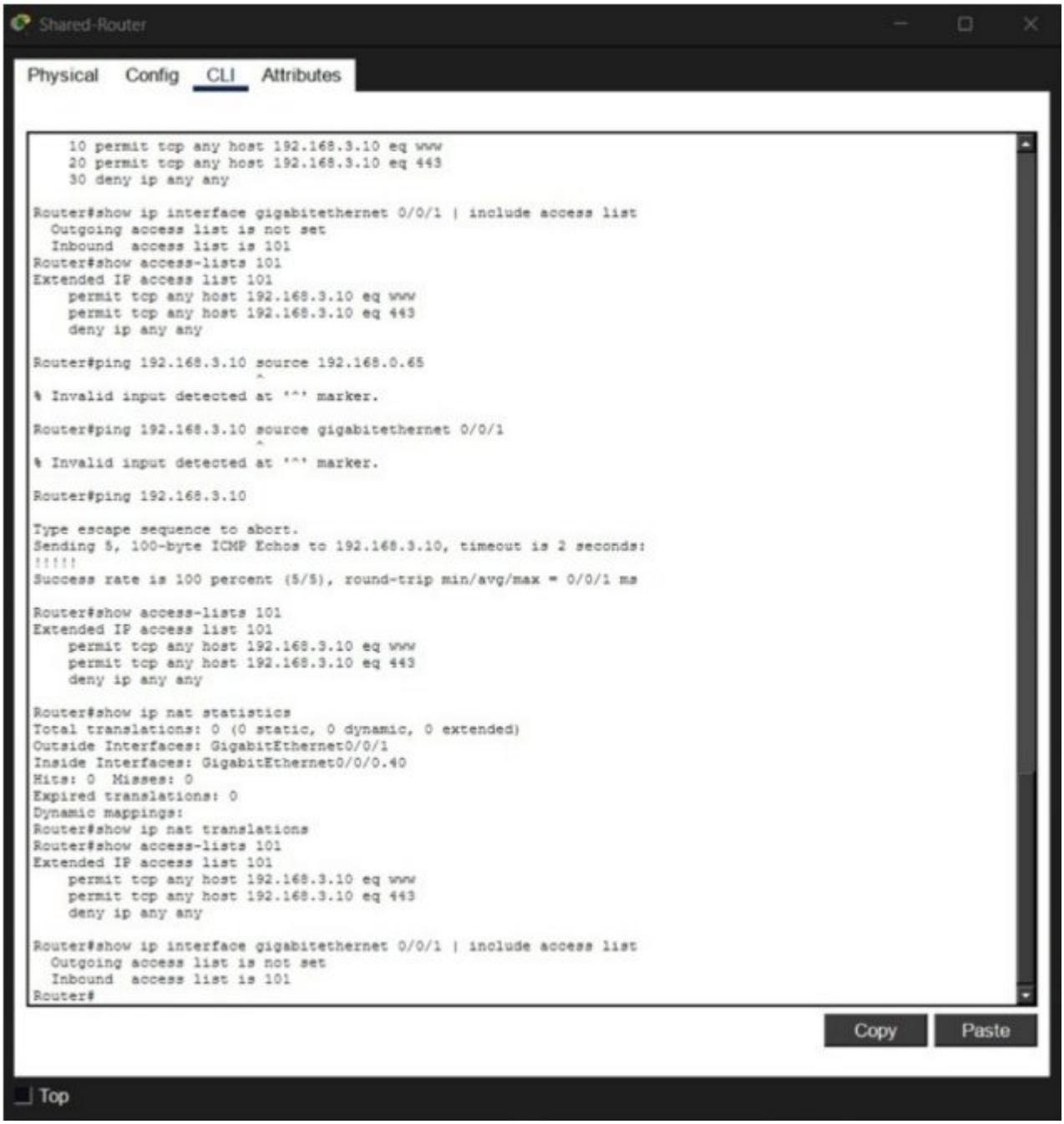
The above screenshot is the continued NAT and ACL configurations.



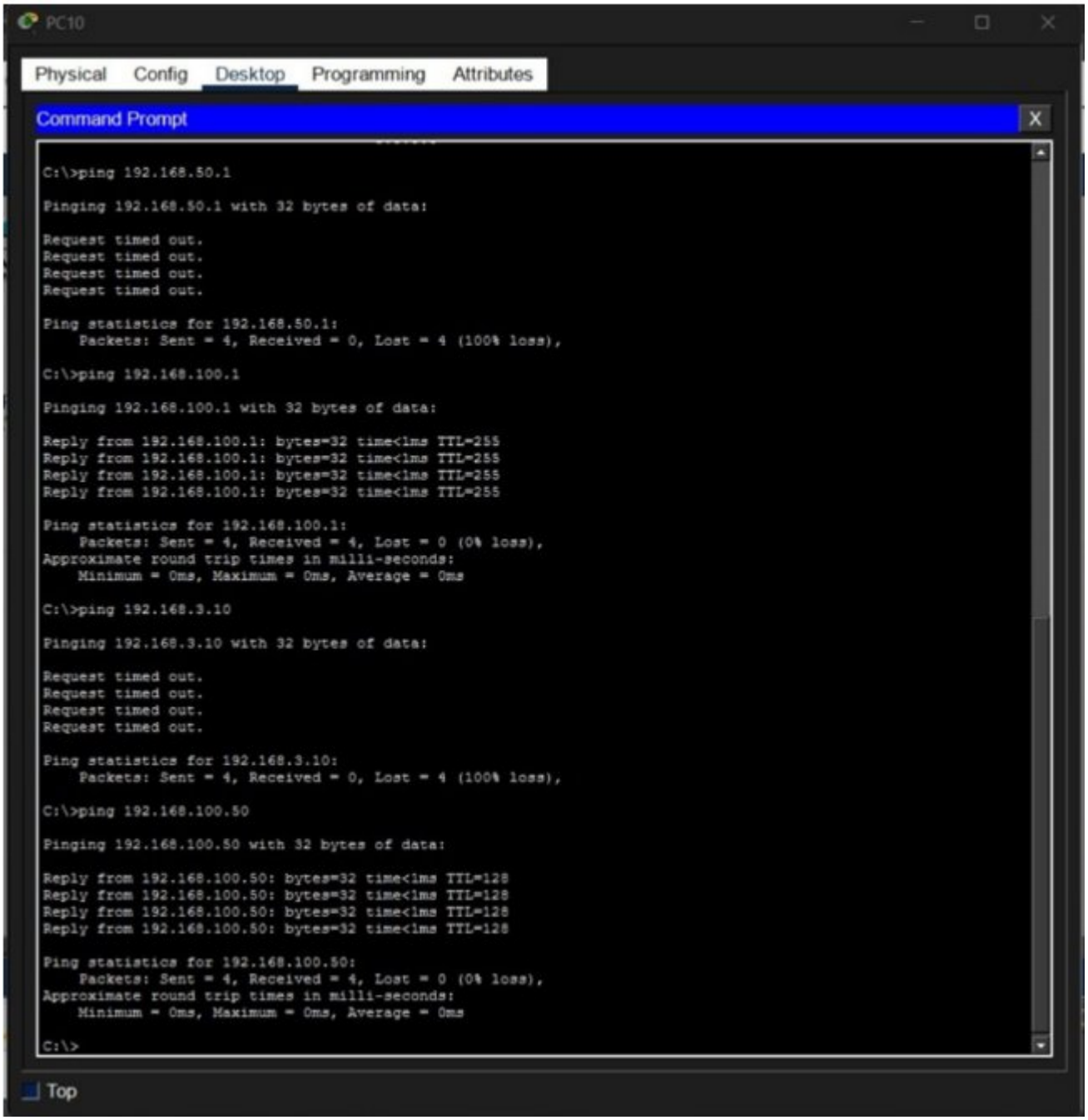
The above screenshot shows both the NAT setup and ACL rules applied to restrict traffic. It also only allows HTTP (port80) and HTTPS (port 443) traffic to the staging server and blocks all other traffic. The command “show ip nat statistics” was to verify that NAT was active and shows inside/outside interfaces assigned. The command “show access-lists” showed the permit ip address for HTTP/HTTPS and deny for all the others.



The ping test proved that the staging server is reachable when the permitted protocol for traffic is used. Only HTTP/HTTPS traffic is allowed to the staging server.



In the above screenshot, it shows that NAT is active and with inside and outside interfaces assigned. It also shows that the ACL is properly configured and active.



The above image shows the ACL and NAT are properly configured as ping tests were executed. The ping tests gives us clear evidence of which destinations are reachable and which are blocked. The developers and external partner pc was pinged and the results came back with replies and we tried to ping the other teams but instead got request timed out.

3.2) The placement of the staging server should be in a dedicated VLAN/DMZ segment which ensures the isolation from the different departments whilst only allowing access to the developers

and external partners. The DMZ's placement is essential as it reduces the risk of compromises because if the staging server is attacked then the damage would not spread and would stay within the DMZ.

The required control plane policies such as firewall rules would allow inbound traffic using HTTP/HTTPS from the external partners to access the staging server. It denies all other inbound traffic. Furthermore, the ACLs are applied at the router interfaces to apply least privilege. The developers are allowed to update the staging server but other VLANs are denied access. The routing considerations should include the use of OSPF/RIP for internal routing only where it is needed. Also, it should prevent unnecessary routing of traffic into sensitive departments.

The required data plane policies such as segmentation would allow for the separation of the different teams and the staging environment by VLANs to prevent both from interfering with each other. NAT forwarding would also allow for the external partners to connect to the staging server using the router's public IP address as the NAT would hide the internal IP address and thus adding more protection. Additionally, the inter-VLAN routing can also be used by only allowing traffic that is necessary such as the developers updating the staging server. This ensures that other teams cannot access it.

The supporting services which include the DNS resolution approach in which the internal DNS helps developers to find the staging server whereas the external DNS records allow external partners to reach the site. Moreover, the staging server should make use of the HTTPS certificates so that the external partners access is secure and ensure that the certificates are kept updated. Regarding to the logging and monitoring, it is important that a central server is used to collect logs from the routers, switches and the staging server and be checked regularly for suspicious activity. The basic intrusion detection and intrusion prevention system should also be implemented to monitor the staging VLAN for suspicious traffic such as brute force attacks and alerts should be sent to administrators for unusual activities.

Least privilege should be used so that developers can update the staging server, and the external partners can only access the staging server through the HTTP/HTTPS protocols while other team members are blocked from the staging server. Limiting the damage by using DMZs so that if the staging server is compromised then it wouldn't spread throughout the network.

Question 4

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.0.64 0.0.0.31 area 0
Router(config-router)#network 192.168.0.96 0.0.0.3 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Here is the **Shared Routers** OSPF configurations that allow it to find the routing to the rest of the network.

```
Router#enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.0.0
Router(config-router)#end
```

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 192.168.0.96 0.0.0.3 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Here is the **Business Analyst** Router's OSPF and RIP configurations, the Business Analyst router is being used as the translation point in order for the parts of the network that are using RIP v2 to be allowed to communicate with the parts of the network(the shared router) that are using OSPF.

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#version 2
Router(config-router)#no auto-summary
Router(config-router)#network 192.168.0.0
Router(config-router)#end
Router#
```

Here is the RIP v2 configurations for the **Developer** Router, Since the Business Analyst router is being used as the translation point, the Developer Router can make use of the BA router in order to communicate with the Shared Router in floor 3.

```
C:\>ping 192.168.0.66

Pinging 192.168.0.66 with 32 bytes of data:

Reply from 192.168.0.66: bytes=32 time<1ms TTL=125
Reply from 192.168.0.66: bytes=32 time<1ms TTL=125
Reply from 192.168.0.66: bytes=32 time<1ms TTL=125
Reply from 192.168.0.66: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.0.66:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

As shown previously in the document, The Developer computer is able to ping the Staging web server through the use and configuration of the RIP to OSPF.

This shows that the legacy systems (dev/BA) are able to communicate with the modern-day systems (Shared router). As well as looking at the entire systems topology through the image at the top of the document you can see that there is 2 possible routes the traffic can take to get to the Shared router, this allows that in case of a fault the network can make use of the (top) wire which is reserved as a backup traffic lane. This maintains the connectivity in the network system.

Question 4.2

By using both RIP and OSPF within the same network structure, BrightSoft Solutions may encounter problems and challenges with the 2 protocols. This is because RIP and OSPF use different metrics to find the best pathing. RIP uses distance vectors while OSPF uses a link state protocol

Protocol Compatibility- RIP is used for more legacy (older) systems while OSPF is used more in modern day systems, if the network uses version 1 of RIP the older systems will not be able to communicate with the newer systems as OSPF does not have classes. This creates routing failures in the network. In order to go between RIP and OSPF the metric needs to be converted, This could mean for routers to choose longer paths due to the original path possibly being able to translate the metric between the 2 protocols.
Administrative complexity- In order for both protocols to be in the same infrastructure one or more routers must have both protocols active in order to translate the metrics. Using both also makes troubleshooting issues more difficult as it is harder to understand why traffic is taking a certain path. As well as having both protocols uses more resources, often requiring better hardware than just using one of the protocols.
Route Redistribution- Each protocol has a AD value, however, a network will always prefer the lower value, this results in OSPF always winning and due to this the network may choose the OSPF path even if it is slower and longer than the RIP path. Due to the protocols using different metrics, the network manager must manually seed metric, if this is done incorrectly the route may become unusable. In order to house both protocols in the network, the network manager must make sure to be using RIP version 2 otherwise the traffic going from OSPF to RIP will be lost.
Challenges for failover/backup- In order to make a backup link with the correct AD to ensure that the network does not use that over the primary link takes complex calculations, on top of that it requires complex manual configurations in order to set up. This makes the failover link harder and more expensive to implement.
Using both protocols, the backup link will be less resilient as due to RIP only updating every now and then and not like OSPF that does it when a change occurs, the network may experience downtime before the RIP finds the backup route to continue the networks functionality.
If the backup link is not configured correctly and is not hidden from the network behind the primary link, this could make data travel between both links back and forth which may cause delays in traffic as well as firewall and security issues as the traffic is expected to only use the one link at a time.

These are ways in which BrightSoft Solutions may struggle with the RIP and OSPF protocols in one network infrastructure.